MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, UP/DOWN, BINARY, DIGITAL, KEY CODE BBJ

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic up/down binary counter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.9 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BBJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/166-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through FF shall be performed in any order with the following exceptions. If test conditions A,B, or C are part of a sequence of tests, the order of testing shall be as follows: A,B,C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

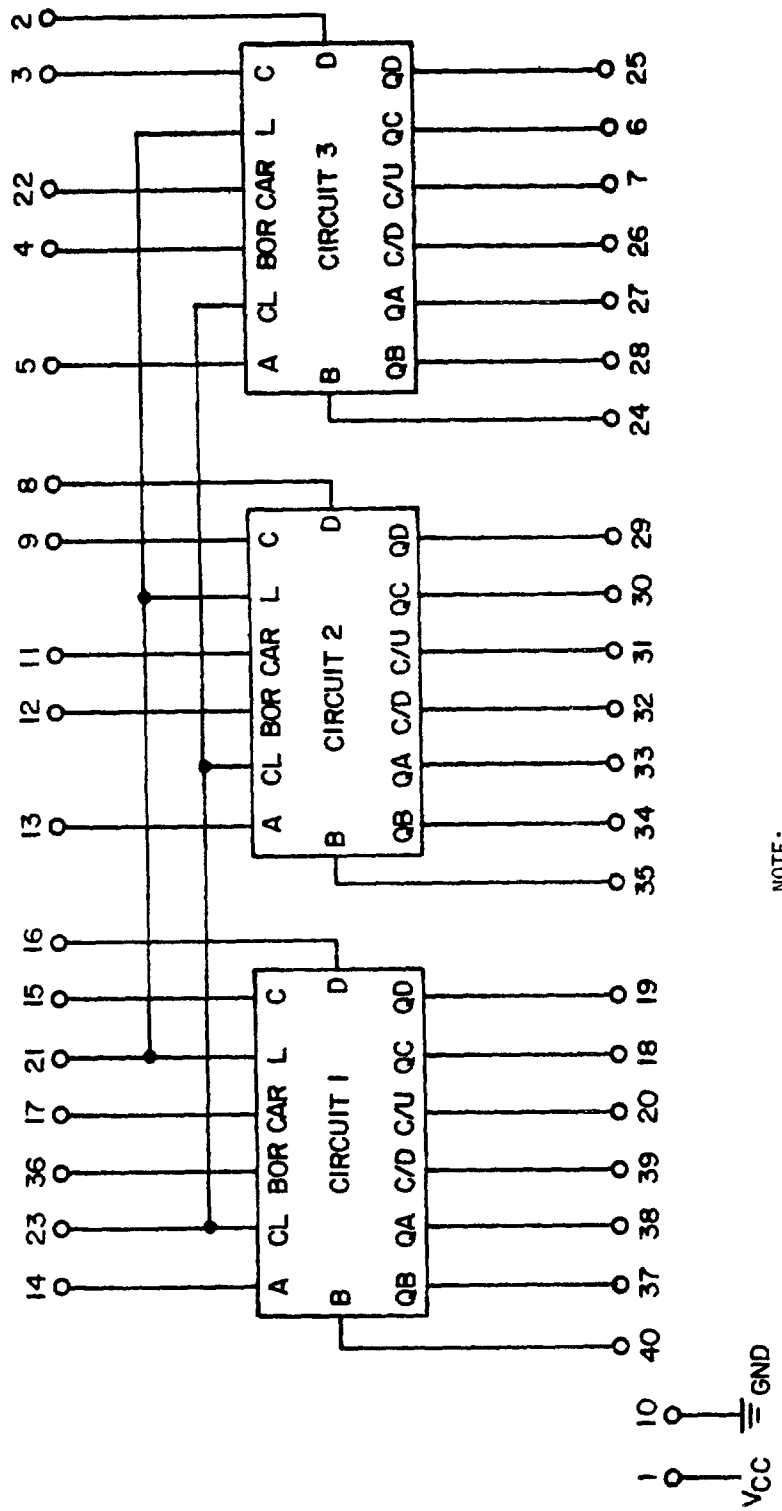
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through FF in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through EE shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through EE shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through FF shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

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NOTE:
A,B,C,D - Data inputs
CL - Clear input
L - Load input
C/D - Count down clock
C/U - Count up clock
QA, QB, QC, QD - Outputs
BOR - Borrow output
CAR - Carry output

FIGURE 1. BBJ module function diagram, up/down binary counter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low loaded)	III	S	50	600	50	600	50	600	50	600	mV
B1-B18	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
B19,B20	Input breakdown current	IV	S	0	300	0	300	0	300	0	300	µA
C	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D20	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E18	Input current, low level	VII	X	10	200	10	200	10	200	10	200	µA
E19,E20	Input current, low level	VII	X	10	600	10	600	10	600	10	600	µA
F1-F18	Input current, high level	VII	X	0	10	0	10	0	10	0	10	µA
F19,F20	Input current, high level	VII	X	0	30	0	30	0	30	0	30	µA
G1-G18	Output short circuit current	VII	X	-3	-15	-3	-15	-3	-15	-3	-15	mA
H1-H18	Output voltage, low level	VIII	X	0	300	0	300	0	300	0	300	mV
J1-J18	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K53	Functional	IX	X	← See Table IX →								
M1,M2	Power supply current, high level and low level	X	X	9	45	9	45	9	45	9	45	mA
N1-N6	Propagation delay time (t_{PHL}), C_L = 50 pF	XI	S	10	130	10	130	10	130	10	130	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N7-N12	Propagation delay time (t_{PHL}), C_L = 150 pF	XI	Q	10	145	10	145	10	145	10	145	ns
P1-P6	Propagation delay time (t_{PLH}), C_L = 50 pF	XI	S	10	130	10	130	10	130	10	130	ns
P7-P12	Propagation delay time (t_{PLH}), C_L = 150 pF	XI	Q	10	145	10	145	10	145	10	145	ns
Q1-Q12	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	10	200	10	200	10	200	10	200	ns
Q13-Q24	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	10	215	10	215	10	215	10	215	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	10	240	10	240	10	240	10	240	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	10	255	10	255	10	255	10	255	ns
S1-S12	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	10	200	10	200	10	200	10	200	ns
S13-S24	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	10	215	10	215	10	215	10	215	ns
T1-T12	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	10	240	10	240	10	240	10	240	ns
T13-T24	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	10	255	10	255	10	255	10	255	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U1-U18	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	10	60	10	60	10	60	10	60	ns
U19-U36	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	10	80	10	80	10	80	10	80	ns
V1-V18	Transition time (t_{THL}), C_L = 50 pF	XIII	S	5	25	5	25	5	25	5	25	ns
V19-V36	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	5	35	5	35	5	35	5	35	ns
W1-W12	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	10	240	10	240	10	240	10	240	ns
W13-W24	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	10	255	10	255	10	255	10	255	ns
X1-X12	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	10	220	10	220	10	220	10	220	ns
X13-X24	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	10	235	10	235	10	235	10	235	ns
Y1-Y12	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	10	200	10	200	10	200	10	200	ns
Y13-Y24	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	10	215	10	215	10	215	10	215	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Z1-Z6	Maximum counting frequency check (carry and borrow period)	XV	S	6.3	6.6	6.3	6.6	6.3	6.6	6.3	6.6	µs
AA1-AA12	Setup time (data to load) and minimum load pulse width check	XVI	S	935	1,070	935	1,070	935	1,070	935	1,070	ns
BB	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
CC1-CC12	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
DD1-DD12	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	mV
EE	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
FF1-FF18	Input capacitance	XX	Q	2	12	2	12	2	12	2	12	pF
FF19,FF20	Input capacitance	XX	Q	6	36	6	36	6	36	6	36	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/166-1 supersedes M28787/166. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5.0 V dc ±10 percent
Maximum power dissipation	248 mW (outputs not loaded)
Clock inputs	
Counting frequency	DC to 2.5 MHz maximum
Pulse width	200 ns minimum
Triggering edge	Positive
Input voltage	
High level	2.0 V minimum, V_{CC} maximum
Low level	-0.5 V minimum, 0.7 V maximum
Input current (data, count/up, count/down) high level	
(V_{IH} = 2.4 V)	10 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
Input current (clear, load) high level	
(V_{IH} = 2.4 V)	30 µA maximum
(V_{IH} = 5.5 V)	300 µA maximum
Input current (data, count/up, count/down) low level	
(V_{IL} = 0.0 V)	0.20 mA maximum
(V_{IL} = 0.3 V)	0.18 mA maximum
Input current (clear, load) low level	
(V_{IL} = 0.0 V)	0.60 mA maximum
(V_{IL} = 0.3 V)	0.54 mA maximum
Output voltage	
High level	2.4 V minimum, V_{CC} maximum
Low level	0.0 V minimum, 0.3 V maximum
Output drive high level	
(V_{OH} = 2.4 V)	-100 µA maximum
Output drive low level	
(V_{OL} = 0.3 V)	1.8 mA maximum
(V_{OL} = 0.0 V)	2.0 mA maximum

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Data, load and clear input pulse width	200 ns minimum
Data setup time (before positive transition of load pulse)	110 ns minimum
Data hold time (after positive transition of load pulse)	0 ns minimum

6.5 Module function. This module consists of three 4-bit synchronous binary up/down counters. Each counter has parallel inputs and outputs and carry and borrow outputs. The counters have common clear and load inputs and independent count/up and count/down clock inputs. The functional diagram is illustrated on figure 1. All three counters are fully programmable; that is, the outputs may be preset to any state by entering the desired data at the DATA inputs while the LOAD input is low. The output will change to agree with the DATA inputs independently of the count pulses. The CLEAR input forces all outputs to the low level when a high level is applied. The CLEAR function is independent of the COUNT and LOAD inputs. Both BORROW and CARRY outputs are available to cascade both the up and down counting functions. The BORROW output produces a pulse equal in width to the COUNT/DOWN input when the counter underflows, and the CARRY output produces a pulse equal in width to the COUNT/UP input when the counter overflows.

6.5.1 Input to output relationship. Each counter has the input to output relationship for positive logic shown in table II.

TABLE II. Input to output relationship. 1/-5/

	Inputs								Outputs			
			Data									
	Clear	Load	A	B	C	D	Count/up	Count/down	Q_A	Q_B	Q_C	Q_D
Asynchronous operation	H	X	X	X	X	X	X	X	L	L	L	L
	L	L	A	B	C	D	X	X	A	B	C	D
Synchronous Operation	L	H	X	X	X	X	H	1/	Count/down			
	L	H	X	X	X	X	1/	H	Count/up			

1/ Function is synchronous with the positive-going edge of the count pulse.

2/ X = Unspecified (either high level or low level).

3/ CARRY = $Q_A \cdot Q_B \cdot Q_C \cdot Q_D \cdot \overline{C/U}$ (CARRY is active low).

4/ BORROW = $\overline{Q}_A \cdot \overline{Q}_B \cdot \overline{Q}_C \cdot \overline{Q}_D \cdot \overline{C/D}$ (BORROW is active low)

5/ L = low level and H = high level.

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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

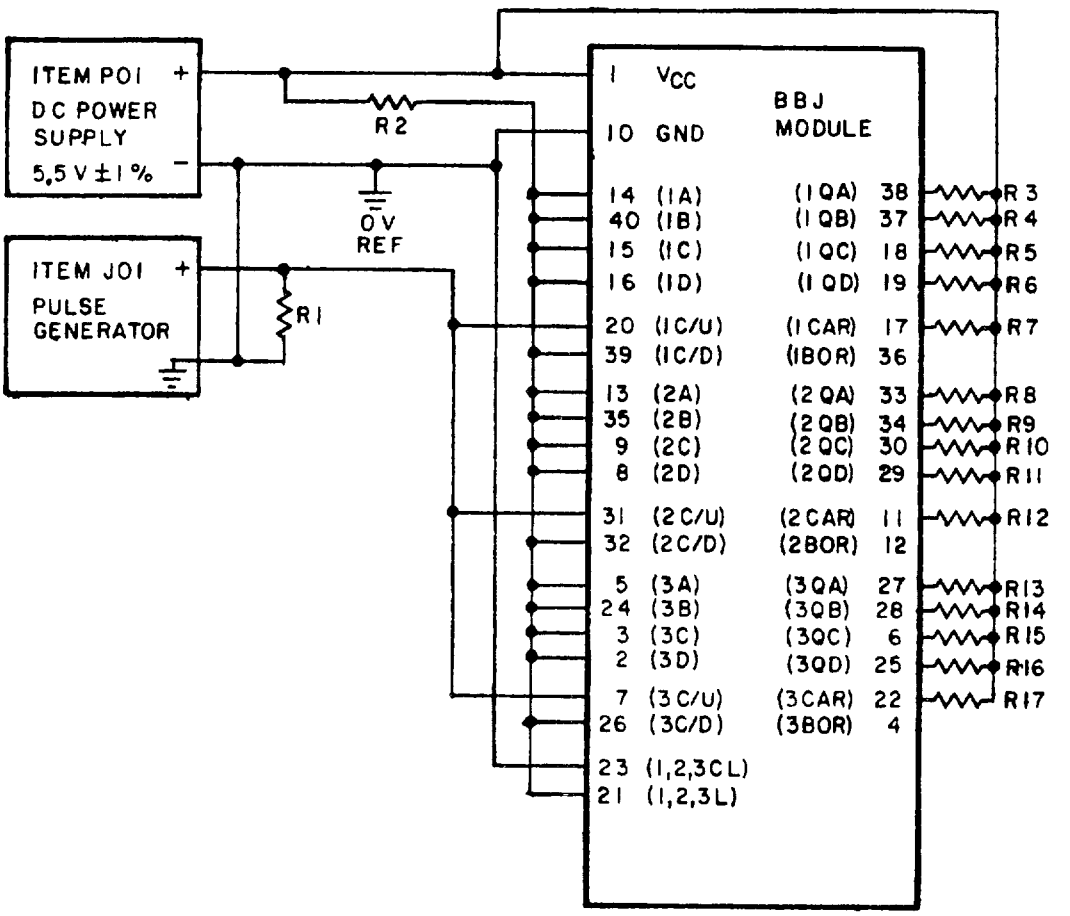
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
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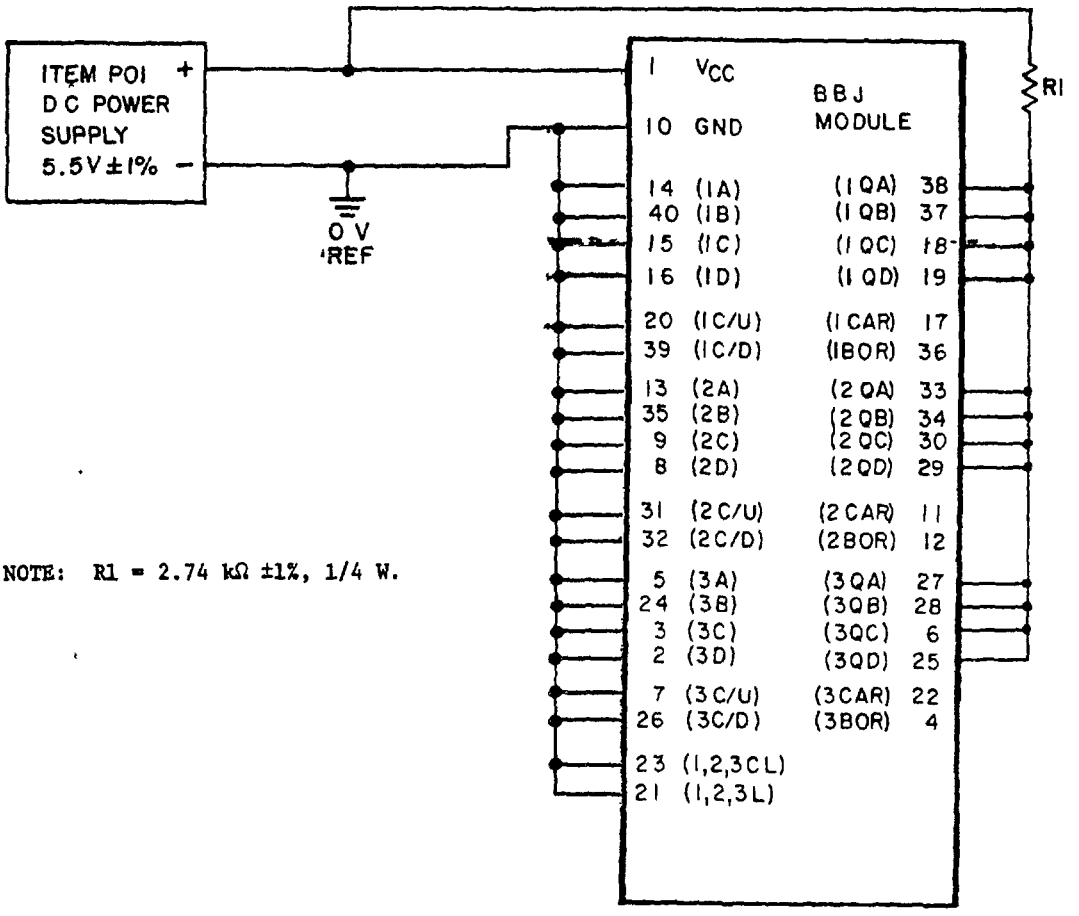
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 1.0 kΩ ±10%, 1/8 W.
3. R3-R17 = 2.74 kΩ ±1%, 1/4 W.
4. Item J01 setup:
   PWA = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 50 ±20 ns.
   PA = 4.0 ±0.2 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1 = 2.74 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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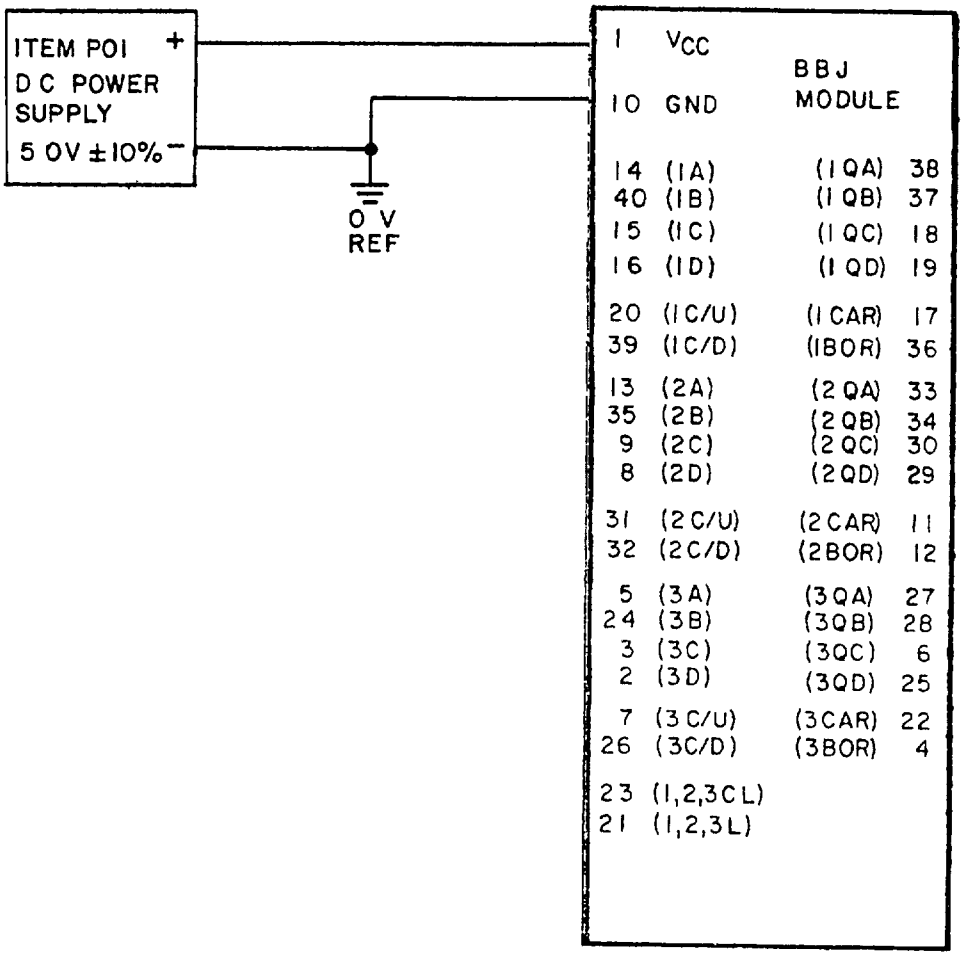


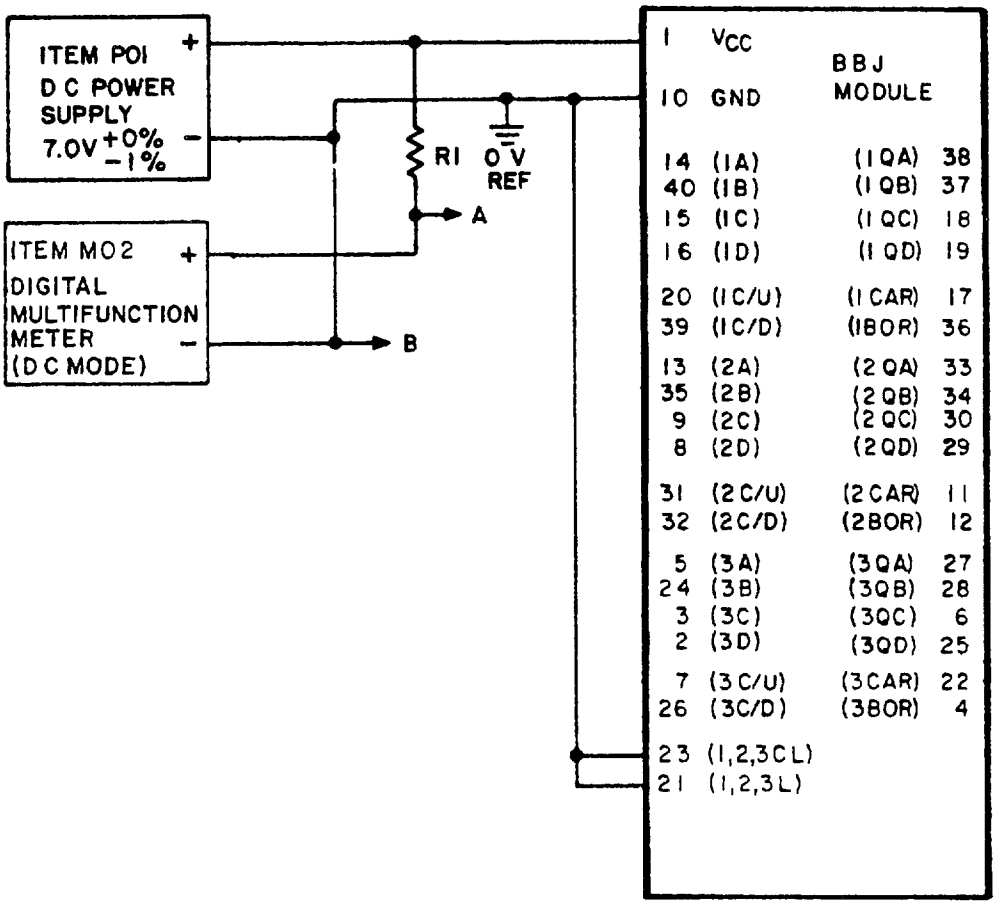
FIGURE 4 Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	1QA V_{OUT} (low loaded)	38	14	50	600	mV	5
A2	1QB	37	40				
A3	1QC	18	15				
A4	1QD	19	16				
A5	1CAR	17	20				
A6	1BOR	36	14,15,16, 39,40				
A7	2QA	33	13				
A8	2QB	34	35				
A9	2QC	30	9				
A10	2QD	29	8				
A11	2CAR	11	31				
A12	2BOR	12	8,9,13, 32,35				
A13	3QA	27	5				
A14	3QB	28	24				
A15	3QC	6	3				
A16	3QD	25	2				
A17	3CAR	22	7				
A18	3BOR	4	2,3,5,24, 26				

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NOTES:
1. R1 = 2.74 kΩ ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pin, monitor for a minimum of 5 seconds before recording measurement.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	1A	Input breakdown current	14	0	100	µA	6
B2	1B		40				
B3	1C		15				
B4	1D		16				
B5	1C/U		20				
B6	1C/D		39				
B7	2A		13				
B8	2B		35				
B9	2C		9				
B10	2D		8				
B11	2C/U		31				
B12	2C/D		32				
B13	3A		5				
B14	3B		24				
B15	3C		3				
B16	3D		2				
B17	3C/U		7				

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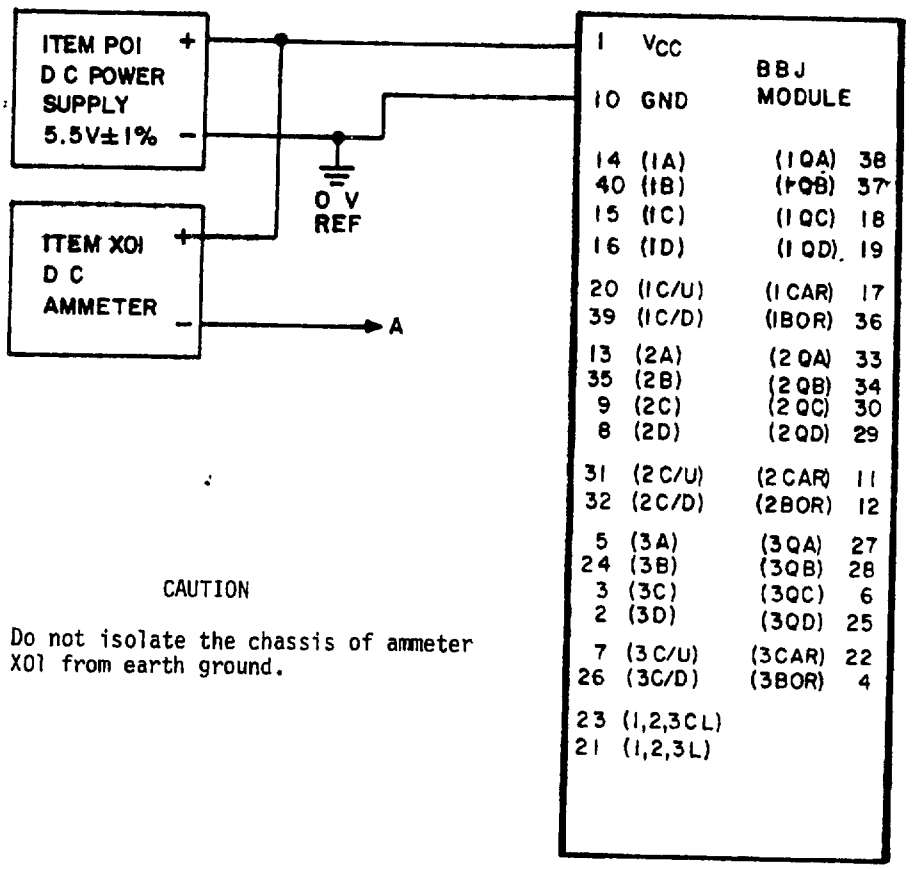
TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
B18	3C/D	Input breakdown current	26	0	100	µA	6
B19	1,2,3CL	Input breakdown current	23	0	300	µA	6
B20	1,2,3L	Input breakdown current	21	0	300	µA	6

TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	All pins	Frame	10	∞	MΩ	7

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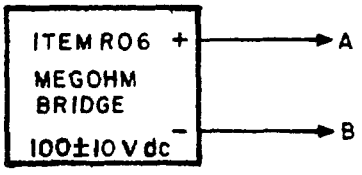


CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

		BBJ MODULE	
1	V_{CC}		
10	GND		
14	(1A)	(1QA)	38
40	(1B)	(1QB)	37
15	(1C)	(1QC)	18
16	(1D)	(1QD)	19
20	(1C/U)	(1CAR)	17
39	(1C/D)	(1BOR)	36
13	(2A)	(2QA)	33
35	(2B)	(2QB)	34
9	(2C)	(2QC)	30
8	(2D)	(2QD)	29
31	(2C/U)	(2CAR)	11
32	(2C/D)	(2BOR)	12
5	(3A)	(3QA)	27
24	(3B)	(3QB)	28
3	(3C)	(3QC)	6
2	(3D)	(3QD)	25
7	(3C/U)	(3CAR)	22
26	(3C/D)	(3BOR)	4
23	(1,2,3CL)		
21	(1,2,3L)		

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

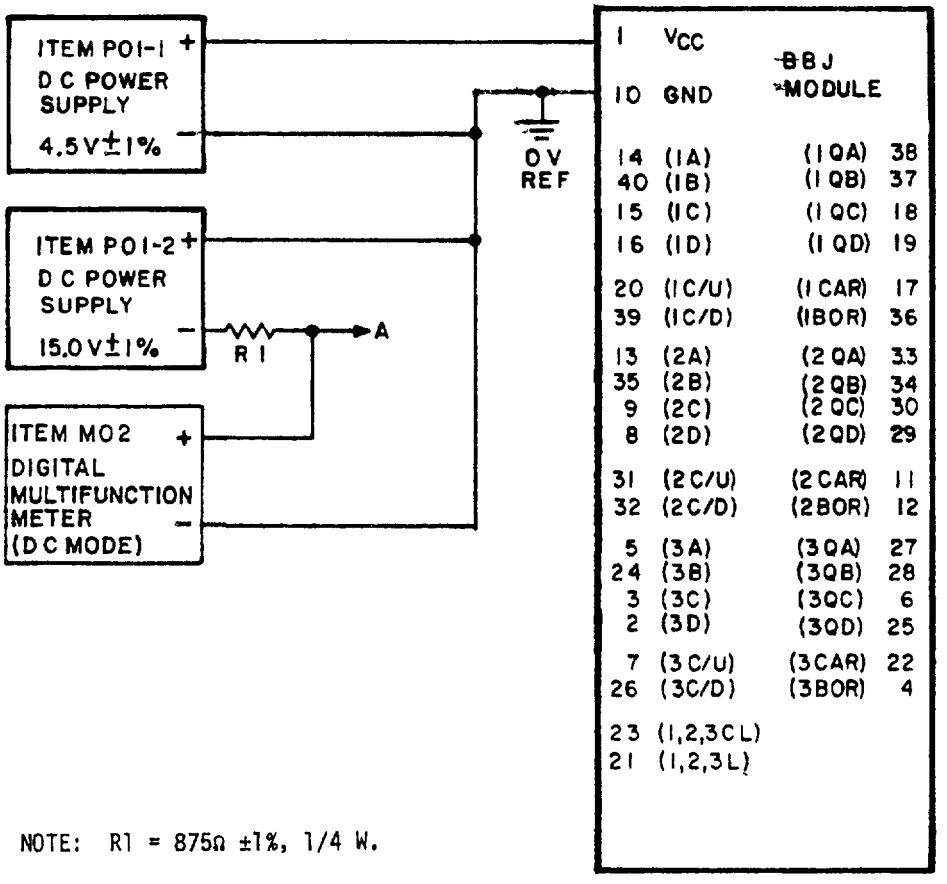
Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	1A	Input clamp diode voltage	14	-0.4	-1.5	V	8
D2	1B		40				
D3	1C		15				
D4	1D		16				
D5	1C/U		20				
D6	1C/D		39				
D7	2A		13				
D8	2B		35				
D9	2C		9				
D10	2D		8				
D11	2C/U		31				
D12	2C/D		32				
D13	3A		5				
D14	3B		24				
D15	3C		3				
D16	3D		2				
D17	3C/U		7				
D18	3C/D		26				

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TABLE VI. Input clamp diode voltage tests - Continued.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D19	1,2,3CL	Input clamp diode voltage	23	-0.4	-1.5	V	8
D20	1,2,3L	Input clamp diode voltage	21	-0.4	-1.5	V	8

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NOTE: R1 = 875Ω ±1%, 1/4 W.

FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1A Input current, low level	21,23	Open	14	Open	10	200	µA	9
E2	1B			40					
E3	1C			15					
E4	1D			16					
E5	1C/U	Open		20					
E6	1C/D	Open		39					
E7	2A	21,23		13					
E8	2B			35					
E9	2C			9					
E10	2D			8					
E11	2C/U	Open		31					
E12	2C/D	Open		32					
E13	3A	21,23		5					
E14	3B			24					
E15	3C			3					
E16	3D			2					
E17	3C/U	Open		7					
E18	3C/D	Open		26					

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
E19	1,2,3CL	Input current, low level	Open	Open	23	Open	10	600	µA	9
E20	1,2,3L	Input current, low level	Open	Open	21	Open	10	600		
F1	1A	Input current, high level	15,16, 20,39, 40	14	Open	21,23	0	10		
F2	1B		14,15, 16,20, 30	40						
F3	1C		14,16, 20,39, 40	15						
F4	1D		14,15, 20,39, 40	16						
F5	1C/U		14,15, 16,21, 23,39, 40	20		Open				
F6	1C/D		14,15, 16,20, 21,23, 40	39		Open				
F7	2A		8,9,31, 32,35	13		21,23				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
F8	2B	Input current, high level	8,9,13, 31.32	35	Open	21,23	0	10	µA	9
F9	2C		8,13, 31,32, 35	9		21,23				
F10	2D		9,13, 31,32, 35	8		21,23				
F11	2C/U		8,9, 13,21, 23,32, 35	31		Open				
F12	2C/D		8,9, 13,21, 23,31, 35	32		Open				
F13	3A		2,3,7, 24,26	5		21,23				
F14	3B		2,3,5, 7,26	24						
F15	3C		2,5,7, 24,26	3						
F16	3D		3,5,7, 24,26	2						
F17	3C/U		2,3,5, 21,23, 24,26	7		Open				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
F18	3C/D	Input current, high level	2,3,5, 7,21, 23,24	26	Open	Open	0	10	µA	9
F19	1,2,3CL	Input current, high level	2,3,5, 7,8,9, 13,14, 15,16, 20,21, 24,26, 31,32, 35,39, 40	23	Open	Open	0	30	µA	↓
F20	1,2,3L	Input current, high level	2,3,5, 7,8,9, 13,14, 15,16, 20,23, 24,26, 31,32, 35,39, 40	21	Open	Open	0	30	µA	↓
G1	1QA	Output short circuit current	21,23	Open	38	39,14, 20	-3	-15	mA	↓
G2	1QB	↓	↓	↓	37	20,40, 39	↓	↓	↓	↓
G3	1QC	↓	↓	↓	18	20,15, 39	↓	↓	↓	↓
G4	1QD	↓	↓	↓	19	20,16, 39	↓	↓	↓	↓

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

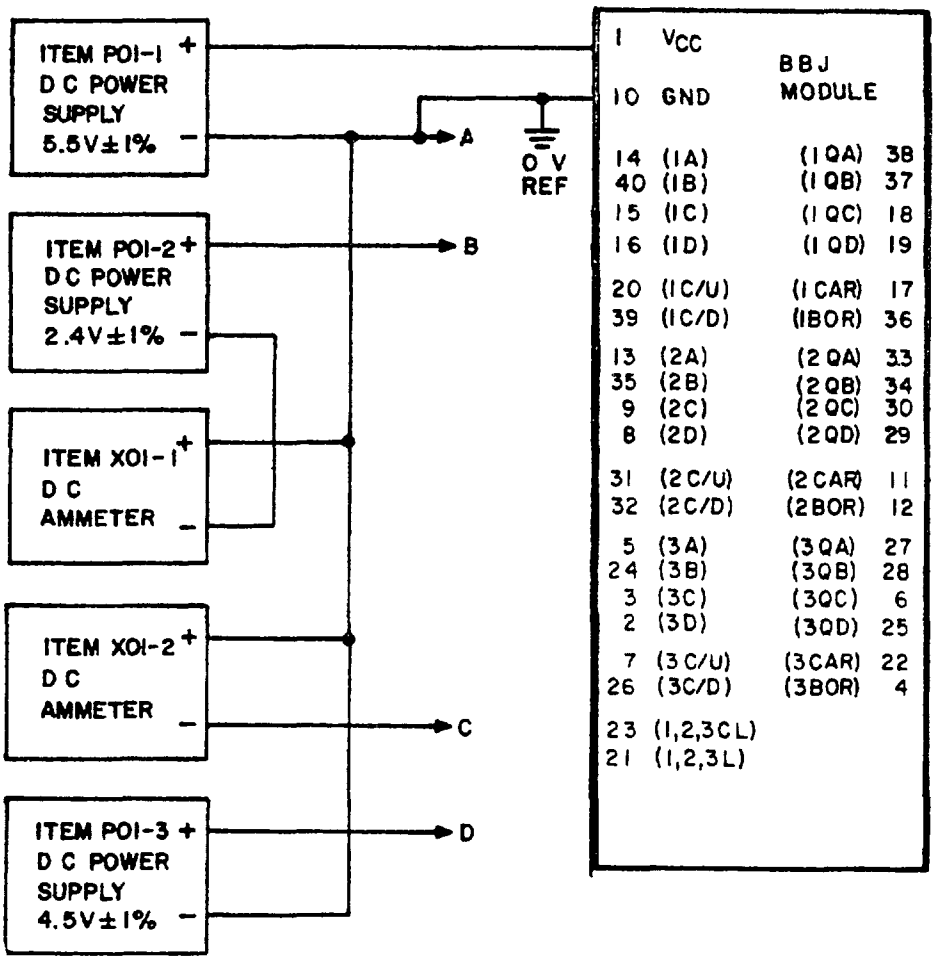
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G5	1CAR	Output short circuit current	21,23	Open	17	20,39	-3	-15	mA	9
G6	1BOR				36	39,20				
G7	2QA				33	31,13, 32				
G8	2QB				34	31,35, 32				
G9	2QC				30	31,9, 32				
G10	2QD				29	31,8, 32				
G11	2CAR				11	31,32				
G12	2BOR				12	32,31				
G13	3QA				27	26,5, 7				
G14	3QB				28	7,24, 26				
G15	3QC				6	7,3, 26				
G15	3QD				25	7,2, 26				

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G17	3CAR	Output short circuit current	21,23	Open	22	7,26	-3	-15	mA	9
G18	3BOR	Output short circuit current	21,23	Open	4	7,26	-3	-15	mA	9

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. Since items XOI-1 and XOI-2 are not used simultaneously, only one item XOI is required.
2. Only one output shall be shorted at a time, and the measurement taken within 10 seconds.

FIGURE 9. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Output logic states: 1QA	1QB	1QC	1QD	1CAR	1BOR	2QA	2QB	2QC	2QD	2CAR	2BOR	3QA	3QB	3QC	3QD	3CAR	3BOR	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
H1	1QA Output voltage, low level	2,3,5,8,9, 13,14,15, 16,21,23, 24,26,32, 35,39,40	7,20,31	38	Open	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	0	300	mV	10
H2	1QB			37																							
H3	1QC			18																							
H4	1QD			19																							
H5	2QA			33																							
H6	2QB			34																							
H7	2QC			30																							
H8	2QD			29																							
H9	3QA			27																							
H10	3QB			28																							
H11	3QC			6																							
H12	3QD			25																							
H13	1BOR			36																							
H14	2BOR			12																							
H15	3BOR			4																							

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TABLE VIII. Output voltage (low-level and high level) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Output logic states: 1QA	1QB	1QC	1QD	1CAR	1BOR	2QA	2QB	2QC	2QD	2CAR	2BOR	3QA	3QB	3QC	3QD	3CAR	3BOR	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
H16	1CAR	Output voltage, low level	7,20,21, 23,31	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40	17	Open	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	0	300	mV	10
H17	2CAR	Output voltage, low level	7,20,21, 23,31	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40	11						L						L						L		0	300	mV	
H18	3CAR	Output voltage, low level	7,20,21, 23,31	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40	22						L						L						L		0	300	mV	
J1	1QA	Output voltage, high level	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	38						H						H						H		2.4	4.5	V	
J2	1QB				37																							
J3	1QC				18																							
J4	1QD				19																							

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TABLE VIII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 C A R	1 B O R	2 Q A	2 Q B	2 Q C	2 Q D	2 C A R	2 B O R	3 Q A	3 Q B	3 Q C	3 Q D	3 C A R	3 B O R	Min	Max		
J5	2QA Output voltage, high level	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	33	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	2.4	4.5	V	10
J6	2QB			34																							
J7	2QC			30																							
J8	2QD			29																							
J9	3QA			27																							
J10	3QB			28																							
J11	3QC			6																							
J12	3QD			25																							
J13	1BOR			36																							
J14	2BOR			12																							
J15	3BOR			4																							
J16	1CAR			17																							
J17	2CAR			11																							
J18	3CAR			22																							

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TABLE IX. Functional test.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 C A R	1 B O R	2 Q A	2 Q B	2 Q C	2 Q D	2 C A R	2 B O R	3 Q A	3 Q B	3 Q C	3 Q D	3 C A R	3 B O R	Min	Max		
K1	Functional, parallel load	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	2/ 3/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	1/			10
K2	Functional, clear	21	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40		Open	L	L	L	L	H		L	L	L	L	H		L	L	L	L	H					
K3	Functional, parallel load	21,23	2,3,5,8, 9,13,14, 15,16,24, 26,32,39, 40,35		7,20, 31	H	H	H	H	L		H	H	H	H	L		H	H	H	H	L					
K4		3,5, 9,13,14, 15,16,21, 23,24,35, 40	2,7,8,16, 20,31		26, 32, 39	L	L	L	H	H		L	L	L	H	H		L	L	L	H	H					
K5		2,8,16, 21,23	3,5,9, 13,14,15, 24,26,32, 35,39,40		7,20 31	H	H	H	L			H	H	H	L			H	H	H	L						
K6		3,9,15, 21,23,24, 35,40	2,5,7,8, 13,14,16, 20,31		26,32, 39	H	L	L	H			H	L	L	H			H	L	L	H						
K7		2,5,8,13, 14,16,21, 23	3,9,15, 24,26,32, 35,39,40		7,20 31	L	H	H	L			L	H	H	L			L	H	H	L						

See footnotes at end of table.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 IQ A	1 IQ B	1 IQ C	1 IQ D	1 IC A R	1 IB O R	2 IQ A	2 IQ B	2 IQ C	2 IQ D	2 IC A R	2 IB O R	3 IQ A	3 IQ B	3 IQ C	3 IQ D	3 IC A R	3 IB O R	Min	Max		
K8	Functional, parallel load	21,23,24, 35,40	2,3,5,7, 8,9,13, 14,15,16, 20,31	2/ 3/	26, 32, 39	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	1/			10
K9		2,3,5,8,9, 13,14,15, 16,21,23	24,26,32, 35,39,40		7,20, 31	L	H	L	L			L	H	L	L			L	H	L	L						
K10		5,13,14, 21,23,24, 35,40	2,3,7,8, 9,15,16, 20,31		26, 32, 39	L	L	H	H			L	L	H	H			L	L	H	H						
K11		2,3,8,9, 15,16,21, 23,24,35, 40	5,13,14, 26,32,39		7,20, 31	H	L	L	L			H	L	L	L			H	L	L	L						
K12		5,13,14, 21,23	2,3,7,8, 9,15,16, 20,24,31, 35,40		26, 32 39	L	H	H	H			L	H	H	H			L	H	H	H						
K13		2,3,8,9, 15,16,21, 23	5,13,14, 24,26,32, 35,39,40		7,20, 31	H	H	L	L			H	H	L	L			H	H	L	L						
K14		2,5,8,13, 14,16,21, 23,24,35, 40	3,7,9,15, 20,31		26, 32, 39	L	L	H	L			L	L	H	L			L	L	H	L						
K15		3,9,15, 21,23	2,5,8,13, 14,16,24, 26,32,35, 39,40		7,20, 31	H	H	L	H			H	H	L	H			H	H	L	H						

See footnotes at end of table.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 IQ A	1 IQ B	1 IQ C	1 IQ D	1 IC A R	1 IB O R	2 IQ A	2 IQ B	2 IQ C	2 IQ D	2 IC A R	2 IB O R	3 IQ A	3 IQ B	3 IQ C	3 IQ D	3 IC A R	3 IB O R	Min	Max		
K16	Functional, parallel load	2,8,16,21, 23,24,35, 40	3,5,7,9, 13,14,15, 20,31	2/ 3/	25, 32, 39	H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	1/			10
K17		3,5,9, 13,14,15, 21,23	2,8,16, 24,26,32, 35,39,40		7,20, 31	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H				
K18		7,20,21, 23,31	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40		Open	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H				
K19		2,3,5,8,9, 13,14,15, 16,21,23, 24,26,32, 35,39,40	7,20,31		Open	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L				
K20		2,3,5,8,9, 13,14,15, 16,21,23, 24,25,40	7,20,31		26, 32, 39	L	L				L	L	L				L	L	L	L			L				
K21	Functional, count/up	2,3,5,8,9, 13,14,15, 16,23,24, 35,40	21,26,32, 39		7,20, 31	H	L				H	H	L				H	H	L	L			H				
K22						L	H					L	H					L	H	L							
K23						H	H					H	H					H	H	L							
K24						L	L	H				L	L	H				L	L	H							
K25						H	L	H				H	L	H				H	L	H							

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 C A R	1 B O R	2 Q A	2 Q B	2 Q C	2 Q D	2 C A R	2 B O R	3 Q A	3 Q B	3 Q C	3 Q D	3 C A R	3 B O R	Min	Max		
K26	Functional, count/up	2,3,5,8,9, 13,14,15, 16,23,24, 35,40	21,26,32, 39	2/ 3/	7,20, 31	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H		1/		10
K27						H	H	H	L			H	H	H	L			H	H	H	L						
K28						L	L	L	H			L	L	L	H			L	L	L	H						
K29						H	L					H	L					H	L								
K30						L	H					L	H					L	H								
K31						H	H					H	H					H	H								
K32						L	L	H				L	L	H				L	L	H							
K33						H	L					H	L					H	L								
K34						L	H					L	H					L	H								
K35						H	H					H	H					H	H								
K36		23	2,3,5,8, 9,13,14, 15,16,21, 24,26,32, 35,39,40			L	L	L	L	H		L	L	L	L	H		L	L	L	L	H					
K37	Functional, parallel load	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,31, 35,40		26, 32, 39	H	H	H	H	H		H	H	H	H	H		H	H	H	H	H					

See footnotes at end of table.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 C A R	1 B O R	2 Q A	2 Q B	2 Q C	2 Q D	2 C A R	2 B O R	3 Q A	3 Q B	3 Q C	3 Q D	3 C A R	3 B O R	Min	Max		
K38	Functional, count/down	23	2,3,5,7, 8,9,13, 14,15,16, 20,21,24, 31,35,40	2/ 3/	26, 32, 39	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	1/			10
K39						H	L	H				H	L	H				H	L	H							
K40						L	L	H				L	L	H				L	L	H							
K41						H	H	L				H	H	L				H	H	L							
K42						L	H					L	H					L	H								
K43						H	L					H	L					H	L								
K44						L	L					L	L					L	L								
K45						H	H	H	L			H	H	H	L			H	H	H	L						
K46						L	H					L	H					L	H								
K47						H	L					H	L					H	L								
K48						L	L					L	L					L	L								
K49						H	H	L				H	H	L				H	H	L							
K50						L	H					L	H					L	L								
K51						H	L					H	L					H	L								
K52	↓	↓	↓	↓	↓	L	L					L	L					L	L					↓	↓		↓

See footnotes at end of page.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 C A R	1 B O R	2 Q A	2 Q B	2 Q C	2 Q D	2 C A R	2 B O R	3 Q A	3 Q B	3 Q C	3 Q D	3 C A R	3 B O R	Min	Max		
K53	Functional, count/down	23	2,3,5,7, 8,9,13, 14,15,16, 20,21,24, 31,35,40	2/ 3/	26, 32, 39	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	1/			10

1/ Output logic states:
L = Low-level = 0.0 to 0.3 V.
H = High level = 2.4 to 4.5 V.

2/ Four test conditions K1-K53, verification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after the input conditions have been accomplished. Test conditions K1-K53 are to be performed in sequence.

3/ 1QA, 1QB, 1QC, 1QD, 1CAR, 1BOR, 2QA, 2QB, 2QC, 2QD, 2CAR, 2BOR, 3QA, 3QB, 3QC, 3QD, 3CAR, and 3BOR output logic states are measured at output pins 38, 37, 18, 19, 17, 36, 33, 34, 30, 29, 11, 12, 27, 28, 6, 25, 22, and 4 respectively.

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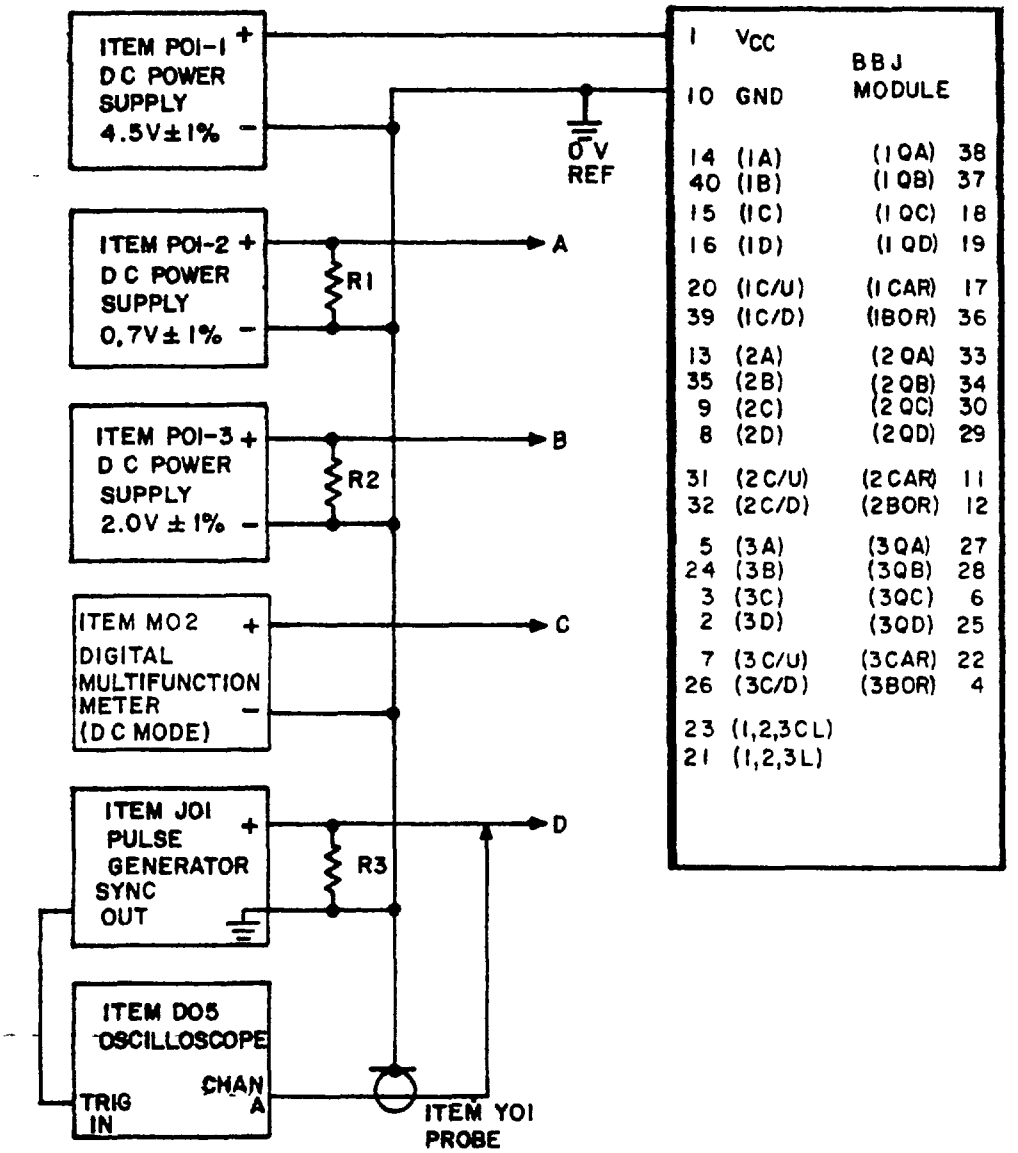
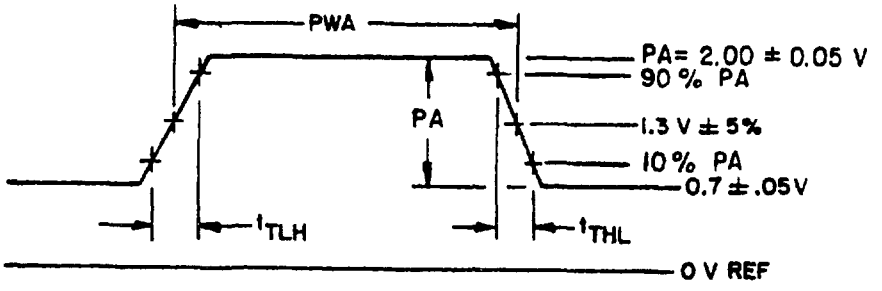


FIGURE 10. Output voltage (low-level and high-level) and functional test circuit for tables VIII and IX.

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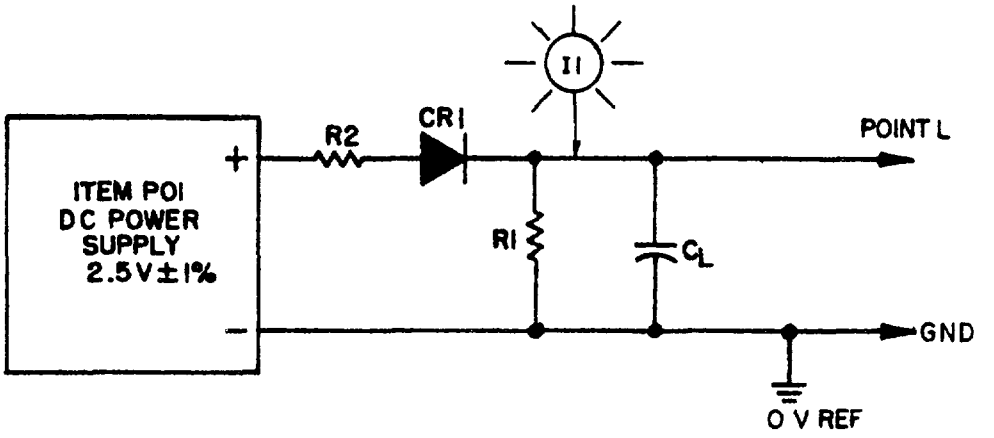
NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. For test conditions H1-H18 and J1-J18, load each output of the circuit under test with the load as specified on figure 11, notes 1 and 3.
3. For test conditions K1-K53, loading of output is not required.
4. For test conditions K1-K53:
   After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply one pulse before making each measurement. Test conditions K1-K53 are to be performed in sequence.
5. Item J01 setup:



PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 50 ±25 ns.
Place item J01 in the single pulse mode.

FIGURE 10. Output voltage (low-level and high-level) and functional test circuit for tables VIII and IX - Continued.

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NOTES:
LOAD NO. 1
1. For test conditions H1-H18, J1-J18, N1-N6, P1-P6, Q1-Q12, R1-R12, S1-S12, T1-T12, U1-U18, V1-V18, W1-W12, X1-X12, Y1-Y12, Z, CC and DD.

   a. CR1 = 1N4150 or equivalent.
   b. C_L = 50 pF ±5% (including probe and parasitic capacitance).
   c. R1 = 24.3 kΩ ±1%, 1/8 W.
   d. R2 = Select value for a current flow of 1.80 mA ±5% out of point L with point L held at 0.3 V.
   e. I1 = Logic state indicator compatible with the logic levels and drive capabilities of the module. (Required for test conditions K and L only.)

LOAD NO. 2
2. For test conditions N7-N12, P7-P12, Q13-Q24, R13-R24, S13-S24, T13-T24, U19-U36, V19-V36, W13-W24, X13-X24, and Y13-Y24.

   a. Same as 1.a.
   b. C_L = 150 pF ±5% (including probe and parasitic capacitance).
   c. Same as 1.c.
   d. Same as 1.d.
   e. Same as 1.e.

3. Capacitor C_L is not required for test conditions H, J, K, and L.

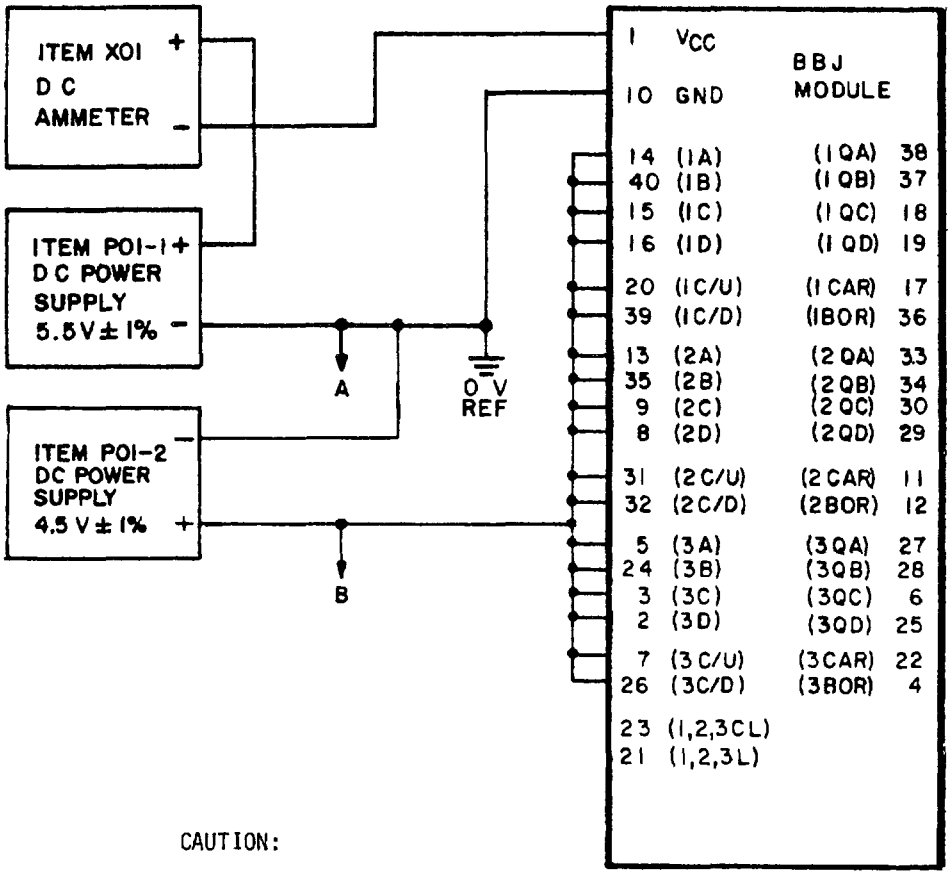
FIGURE 11. BBJ logic load circuit.

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TABLE X. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, high level	21,23	Open	9	45	mA	12
M2	Power supply current, low level	Open	23	9	45	mA	12

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

FIGURE 12. Power supply current (high-level and low-level) test circuit for table X.

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TABLE XI. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
N1	1C/U to 1CAR	Propagation delay time (t_{PHL}), C_L = 50 pF	14	15,16, 40,39	20	21	20	17	10	130	ns	13,14
N2	2C/U to 2CAR		13	9,8, 35,32	31		31	11				
N3	3C/U to 3CAR		5	3,2, 24,26	7		7	22				
N4	1C/D to 1BOR		15,16, 40	14,20	39		39	36				
N5	2C/D to 2BOR		8,9, 35	13,31	32		32	12				
N6	3C/D to 3BOR		2,3, 24	5,7	26		26	4				
N7	1C/U to 1CAR	Propagation delay time (t_{PHL}), C_L = 150 pF	14	15,16 40,39	20		20	17		145		
N8	2C/U to 2CAR	Propagation delay time (t_{PHL}), C_L = 150 pF	13	9,8, 35,32	31		31	11		145		

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TABLE XI. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
N9	3C/U to 3CAR	Propagation delay time (t_{PHL}), C_L = 150 pF	5	3,2, 24,26	7	21	7	22	10	145	ns	13,14
N10	1C/D to 1BOR	↓	15,16, 40	14,20	39	↓	39	36	↓	↓	↓	↓
N11	2C/D to 2BOR	↓	8,9, 35	13,31	32	↓	32	12	↓	↓	↓	↓
N12	3C/D to 3BOR	↓	2,3, 24	5,7	26	↓	26	4	↓	↓	↓	↓
P1	1C/U to 1CAR	Propagation delay time (t_{PLH}), C_L = 50 pF	14	15,16, 40,39	20	↓	20	17	↓	130	↓	↓
P2	2C/U to 2CAR	↓	13	9,8, 35,32	31	↓	31	11	↓	↓	↓	↓
P3	3C/U to 3CAR	↓	5	3,2, 24,26	7	↓	7	22	↓	↓	↓	↓
P4	1C/D to 1BOR	↓	15,16, 40	14,20	39	↓	39	36	↓	↓	↓	↓

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TABLE XI. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
P5	2C/D to 2BOR	Propagation delay time (t_{PLH}), C_L = 50 pF	8,9, 35	13,31	32	21	32	12	10	130	ns	13,14
P6	3C/D to 3BOR	Propagation delay time (t_{PLH}) C_L = 50 pF	2,3,24	5,7	26		26	4		130		
P7	1C/U to 1CAR	Propagation delay time (t_{PLH}), C_L = 150 pF	14	15,16 40,39	20		20	17		145		
P8	2C/U to 2CAR		13	9,8, 35,32	31		31	11				
P9	3C/U to 3CAR		5	3,2, 24,26	7		7	22				
P10	1C/D to 1BOR		15,16, 40	14,20	39		39	36				
P11	2C/D to 2BOR		8,9, 35	13,31	32		32	12				
P12	3C/D to 3BOR		2,3, 24	5,7	26		26	4				

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q1	1C/U to 1QA Propagation delay time (t_{PLH}), C_L = 50 pF	14,15, 16,40	39	20	21	20	38	10	200	ns	13,15
Q2	2C/U to 2QA	8,9, 13,35	32	31		31	33				
Q3	3C/U to 3QA	2,3, 5,24	26	7		7	27				
Q4	1C/U to 1QB	15,16, 40	14,39	20		20	37				
Q5	1C/U to 2QB	8,9, 35	13,32	31		31	34				
Q6	3C/U to 3QB	2,3, 24	5,26	7		7	28				
Q7	1C/U to 1QC	15,16	14,40, 39	20		20	18				
Q8	1C/U to 2QC	8,9	13,35, 32	31		31	30				

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q9	3C/U to 3QC	Propagation delay time (t_{PLH}), C_L = 50 pF	2,3,	5,24, 26	7	21	7	6	10	200	ns	13,15
Q10	1C/U to 1QD	↓	16	14,15, 40,39	20	↓	20	19	↓	↓	↓	↓
Q11	2C/U to 2QD	↓	8	9,13, 35,32	31	↓	31	29	↓	↓	↓	↓
Q12	3C/U to 3QD	↓	2	3,5, 24,26	7	↓	7	25	↓	↓	↓	↓
Q13	1C/U to 1QA	Propagation delay time (t_{PLH}), C_L = 150 pF	14,15, 16,40	39	20	↓	20	38	↓	215	↓	↓
Q14	2C/U to 2QA	↓	8,9, 13,35	32	31	↓	31	33	↓	↓	↓	↓
Q15	3C/U to 3QA	↓	2,3, 5,24	26	7	↓	7	27	↓	↓	↓	↓
Q16	1C/U to 1QB	↓	15,16, 40	14,39	20	↓	20	37	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q17	2C/U to 2QB	Propagation delay time (t_{PLH}), C_L = 150 pF	8,9, 35	13,32	31	21	31	34	10	215	ns	13,15
Q18	3C/U to 3QB	↓	2,3, 24	5,26	7	↓	7	28	↓	↓	↓	↓
Q19	1C/U to 1QC	↓	15,16	14,40, 39	20	↓	20	18	↓	↓	↓	↓
Q20	2C/U to 2QC	↓	8,9	13,35, 32	31	↓	31	30	↓	↓	↓	↓
Q21	3C/U to 3QC	↓	2,3	5,24, 26	7	↓	7	6	↓	↓	↓	↓
Q22	1C/U to 1QD	↓	16	14,15, 40,39	20	↓	20	19	↓	↓	↓	↓
Q23	2C/U to 2QD	↓	8	9,13, 35,32	31	↓	31	29	↓	↓	↓	↓
Q24	3C/U to 3QD	↓	2	3,5, 26,24	7	↓	7	25	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R1	1C/U to 1QA	Propagation delay time (t_{PHL}), C_L = 50 pF	15,16, 40	14,39	20	21	20	38	10	240	ns	13,15
R2	2C/U to 2QA	↓	8,9, 35	13,32	31	↓	31	33	↓	↓	↓	↓
R3	3C/U to 3QA	↓	2,3, 24	5,26	7	↓	7	27	↓	↓	↓	↓
R4	1C/U to 1QB	↓	15,16	14,40, 39	20	↓	20	37	↓	↓	↓	↓
R5	2C/U to 2QB	↓	8,9	13,35, 32	31	↓	31	34	↓	↓	↓	↓
R6	3C/U to 3QB	↓	2,3	5,24, 26	7	↓	7	28	↓	↓	↓	↓
R7	1C/U to 1QC	↓	16	14,15, 40,39	20	↓	20	18	↓	↓	↓	↓
R8	2C/U to 2QC	↓	8	9,13, 35,32	31	↓	31	30	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R9	3C/U to 3QC	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3,5, 24,26	7	21	7	6	10	240	ns	13,15
R10	1C/U to 1QD	↓	Open	14,16, 15,40, 39	20	↓	20	19	↓	↓	↓	↓
R11	2C/U to 2QD	↓	Open	8,13, 9,35, 32	31	↓	31	29	↓	↓	↓	↓
R12	3C/U to 3QD	↓	Open	2,5,3, 24,26	7	↓	7	25	↓	↓	↓	↓
R13	1C/U to 1QA	Propagation delay time (t_{PHL}), C_L = 150 pF	15,16, 40	14,39	20	↓	20	38	↓	255	↓	↓
R14	2C/U to 2QA	↓	8,9, 35	13,32	31	↓	31	33	↓	↓	↓	↓
R15	3C/U to 3QA	↓	2,3, 24	5,26	7	↓	7	27	↓	↓	↓	↓
R16	1C/U to 1QB	↓	15,16	14,40, 39	20	↓	20	37	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R17	2C/U to 2QB	Propagation delay time (t_{PHL}), C_L = 150 pF	8,9	13,35, 32	31	21	31	34	10	255	ns	13,15
R18	3C/U to 3QB		2,3	5,24, 26	7		7	28				
R19	1C/U to 1QC		16	14,15, 40,39	20		20	18				
R20	2C/U to 2QC		8	9,13, 35,32	31		31	30				
R21	3C/U to 3QC		2	3,5, 24,26	7		7	6				
R22	1C/U to 1QD		Open	14,16, 15,40, 39	20		20	19				
R23	2C/U to 2QD		Open	8,13, 9,35, 32	31		31	29				
R24	3C/U to 3QD		Open	2,5, 3,24, 26	7		7	25				

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S1	1C/D to 1QA	Propagation delay time (t_{PLH}), C_L = 50 pF	14,15, 16	40,20	30	21	39	38	10	200	ns	13,15
S2	2C/D to 2QA	↓	8,9, 13	35,31	32	↓	32	33	↓	↓	↓	↓
S3	3C/D to 3QA	↓	2,3, 5	24,7	26	↓	26	27	↓	↓	↓	↓
S4	1C/D to 1QB	↓	14,16, 40	15,20	39	↓	39	37	↓	↓	↓	↓
S5	2C/D to 2QB	↓	8,13, 35	9,31	32	↓	32	34	↓	↓	↓	↓
S6	3C/D to 3QB	↓	2,5, 24	3,7	26	↓	26	28	↓	↓	↓	↓
S7	1C/D to 1QC	↓	14,15, 40	16,20	39	↓	39	18	↓	↓	↓	↓
S8	2C/D to 2QC	↓	9,13, 35	8,31	32	↓	32	30	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S9	3C/D to 3QC	Propagation delay time (t_{PLH}), C_L = 50 pF	3,5, 24	2,7	26	21	26	6	10	200	ns	13,15
S10	1C/D to 1QD	↓	14,15, 16,40	20	39	↓	39	19	↓	↓	↓	↓
S11	2C/D to 2QD	↓	8,9, 13,35	31	32	↓	32	29	↓	↓	↓	↓
S12	3C/D to 3QD	↓	2,3, 5,24	7	26	↓	26	25	↓	↓	↓	↓
S13	1C/D to 1QA	Propagation delay time (t_{PLH}), C_L = 150 pF	14,15, 16	40,20	39	↓	39	38	↓	215	↓	↓
S14	2C/D to 2QA	↓	8,9, 13	35,31	32	↓	32	33	↓	↓	↓	↓
S15	3C/D to 3QA	↓	2,3, 5	24,7	26	↓	26	27	↓	↓	↓	↓
S16	1C/D to 1QB	↓	14,16, 40	15,20	39	↓	39	37	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S17	2C/D to 2QB	Propagation delay time (t_{PLH}), C_L = 150 pF	8,13, 35	9,31	32	21	32	34	10	215	ns	13,15
S18	3C/D to 3QB	↓	2,5, 24	3,7	26	↓	26	28	↓	↓	↓	↓
S19	1C/D to 1QC		14,15, 40	16,20	39		39	18				
S20	2C/D to 2QC		9,13, 35	8,31	32		32	30				
S21	3C/D to 3QC		3,5, 24	2,7	26		26	6				
S22	1C/D to 1QD		14,15, 16,40	20	39		39	19				
S23	2C/D to 2QD		8,9, 13,35	31	32		32	29				
S24	3C/D to 3QD		2,3, 5,24	7	26		26	25				

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
T1	1C/D to 1QA	Propagation delay time (t_{PHL}), C_L = 50 pF	15,16, 40	14,20	39	21	39	38	10	240	ns	13,15
T2	2C/D to 2QA	↓	8,9, 35	13,31	32	↓	32	33	↓	↓	↓	↓
T3	3C/D to 3QA	↓	2,3, 24	5,7	26	↓	26	27	↓	↓	↓	↓
T4	1C/D to 1QB	↓	14,15, 16	40,20	39	↓	39	37	↓	↓	↓	↓
T5	2C/D to 2QB	↓	8,9, 13	35,31	32	↓	32	34	↓	↓	↓	↓
T6	3C/D to 3QB	↓	2,3, 5	24,7	26	↓	26	28	↓	↓	↓	↓
T7	1C/D to 1QC	↓	14,16, 40	15,20	39	↓	39	18	↓	↓	↓	↓
T8	2C/D to 2QC	↓	8,13, 35	9,31	32	↓	32	30	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
T9	3C/D to 3QC	Propagation delay time (t_{PHL}), C_L = 50 pF	2,5, 24	3,7	26	21	26	6	10	240	ns	13,15
T10	1C/D to 1QD	↓	14,15, 40	16,20	39	↓	39	19	↓	↓	↓	↓
T11	2C/D to 2QD	↓	9,13, 35	8,31	32	↓	32	29	↓	↓	↓	↓
T12	3C/D to 3QD	↓	3,5, 24	2,7	26	↓	26	25	↓	↓	↓	↓
T13	1C/D to 1QA	Propagation delay time (t_{PHL}), C_L = 150 pF	15,16, 40	14,20	39	↓	39	38	↓	255	↓	↓
T14	2C/D to QA	↓	8,9, 35	13,31	32	↓	32	33	↓	↓	↓	↓
T15	3C/D to 3QA	↓	2,3, 24	5,7	26	↓	26	27	↓	↓	↓	↓
T16	1C/D to 1QB	↓	14,15, 16	40,20	39	↓	39	37	↓	↓	↓	↓

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
T17	2C/D to 2QB	Propagation delay time (t_{PHL}), C_L = 150 pF	8,9, 13	35,31	32	21	32	34	10	255	ns	13,15
T18	3C/D to 3QB	↓	2,3, 5	24,7	26	↓	26	28	↓	↓	↓	↓
T19	1C/D to 1QC		14,16, 40	15,20	39		39	18				
T20	2C/D to 2QC		8,13, 35	9,31	32		32	30				
T21	3C/D to 3QC		2,5, 24	3,7	26		26	6				
T22	1C/D to 1QD		14,15, 40	16,20	39		39	19				
T23	2C/D to 2QD		9,13, 35	8,31	32		32	29				
T24	3C/D to 3QD		3,5, 24	2,7	26		26	25				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U1	1CAR	Transition time (t_{TLH}), C_L = 50 pF	14	15,16, 40,39	20	21	20	17	10	60	ns	13,14
U2	2CAR		13	9,8, 35,32	31		31	11				
U3	3CAR		5	3,2, 24,26	7		7	22				
U4	1BOR		15,16, 40	14,20	39		39	36				
U5	2BOR		8,9, 35	13,31	32		32	12				
U6	3BOR		2,3, 24	5,7	26		26	4				
U7	1QA		14,15, 16,40	39	20		20	38				13,15
U8	2QA		8,9, 13,35	32	31		31	33				
U9	3QA		2,3, 5,24	26	7		7	27				
U10	1QB		15,16, 40	14,39	20		20	37				
U11	2QB		8,9, 35	13,32	31		31	34				
U12	3QB		2,3, 21	5,26	7		7	28				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U13	1QC	Transition time (t_{TLH}), C_L = 50 pF	15,16	14,40, 39	20	21	20	18	10	60	ns	13,15
U14	2QC	↓	8,9	13,35, 32	31	↓	31	30	↓	↓	↓	↓
U15	3QC	↓	2,3	5,24, 26	7	↓	7	6	↓	↓	↓	↓
U16	1QD	↓	16	14,15, 40,39	20	↓	20	19	↓	↓	↓	↓
U17	2QD	↓	8	9,13, 35,32	31	↓	31	29	↓	↓	↓	↓
U18	3QD	↓	2	3,5, 24,26	7	↓	7	25	↓	↓	↓	↓
U19	1CAR	Transition time (t_{TLH}), C_L = 150 pF	14	15,16, 40,39	20	↓	20	17	↓	80	↓	13;14
U20	2CAR	↓	13	9,8, 35,32	31	↓	31	11	↓	↓	↓	↓
U21	3CAR	↓	5	3,2, 24,26	7	↓	7	22	↓	↓	↓	↓
U22	1BOR	↓	15,16, 40	14,20	39	↓	39	36	↓	↓	↓	↓
U23	2BOR	↓	8,9, 35	13,31	32	↓	32	12	↓	↓	↓	↓

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U24	3BOR	Transition time (t_{TLH}), C_L = 150 pF	2,3, 24	5,7	26	21	26	4	10	80	ns	13,14
U25	1QA		14,15, 16,40	39	20		20	38				13,15
U26	2QA		8,9, 13,35	32	31		31	33				
U27	3QA		2,3, 5,24	26	7		7	27				
U28	1QB		15,16, 40	14,39	20		20	37				
U29	2QB		8,9, 35	13,32	31		31	34				
U30	3QB		2,3, 24	5,26	7		7	28				
U31	1QC		15,16	14,40, 39	20		20	18				
U32	2QC		8,9	13,35, 32	31		31	30				
U33	3QC		2,3	5,24, 26	7		7	6				
U34	1QD		16	14,15, 40,39	20		20	19				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U35	2QD	Transition time (t_{TLH}), C_L = 150 pF	8	9,13 35,32	31	21	31	29	10	80	ns	13,15
U36	3QD	Transition time (t_{TLH}), C_L = 150 pF	2	3,5 24,26	7		7	25	10	80		13,15
V1	1CAR	Transition time (t_{THL}), C_L = 50 pF	14	15,16, 40,39	20		20	17	5	25		13,14
V2	2CAR		13	8,9, 35,32	31		31	11				
V3	3CAR		5	2,3, 24,26	7		7	22				
V4	1BOR		15,16, 40	14,20	39		39	36				
V5	2BOR		8,9, 35	13,31	32		32	12				
V6	3BOR		2,3, 24	5,7	26		26	4				
V7	1QA		15,16, 40	14,39	20		20	38				13,15
V8	2QA		8,9, 35	13,32	31		31	33				
V9	3QA		2,3, 24	5,26	7		7	27				
V10	1QB		15,16	14,40, 39	20		20	37				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
V11	1QB	Transition time (t_{THL}), C_L = 50 pF	8,9	13,35, 32	31	21	31	34	5	25	ns	13,15
V12	3QB		2,3	5,24, 26	7		7	28				
V13	1QC		16	14,15, 40,39	20		20	18				
V14	2QC		8	9,13, 35,32	31		31	30				
V15	3QC		2	3,5, 24,26	7		7	6				
V16	1QD		Open	14,16, 15,40, 39	20		20	19				
V17	2QD		Open	8,13, 9,35, 32	31		31	29				
V18	3QD		Open	2,5, 3,24, 26	7		7	25				
V19	1CAR	Transition time (t_{THL}), C_L = 150 pF	14	15,16, 40,39	20		20	17		35		13,14
V20	2CAR	Transition time (t_{THL}), C_L = 150 pF	13	8,9 35,32	31		31	11		35		13,14

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

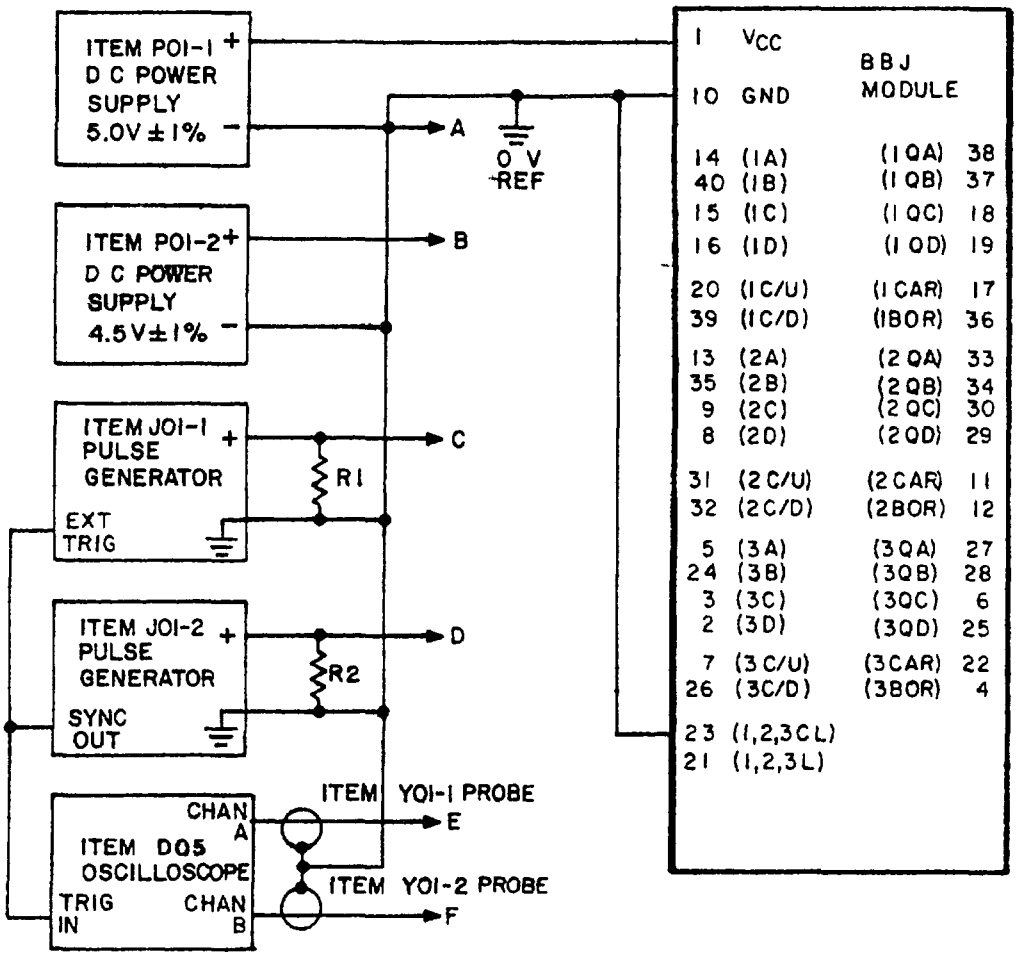
Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
V21	3CAR	Transition time (t_{THL}), C_L = 150 pF	5	2,3, 24,26	7	21	7	22	5	35	ns	13,14
V22	1BOR		15,16, 40	14,20	39		39	36				
V23	2BOR		8,9, 35	13,31	32		32	12				
V24	3BOR		2,3, 24	5,7	26		26	4				
V25	1QA		15,16, 40	14,39	20		20	38				13,15
V26	2QA		8,9, 35	13,32	31		31	33				
V27	3QA		2,3, 24	5,26	7		7	27				
V28	1QB		15,16	14,40, 39	20		20	37				
V29	2QB		8,9	13,35, 32	31		31	34				
V30	3QB		2,3	5,24, 26	7		7	28				
V31	1QC		16	14,15, 40,39	20		20	18				
V32	2QC		8	9,13, 35,32	31		31	30				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
V33	3QC	Transition time (t_{THL}), C_L = 150 pF	2	3,5, 24,26	7	21	7	6	5	35	ns	13,15
V34	1QD	↓	Open	14,16, 15,40, 39	20	↓	20	19	↓	↓	↓	↓
V35	2QD	↓	Open	8,13, 9,35, 32	31	↓	31	29	↓	↓	↓	↓
V36	3QD	↓	Open	2,5, 3,24, 26	7	↓	7	25	↓	↓	↓	↓

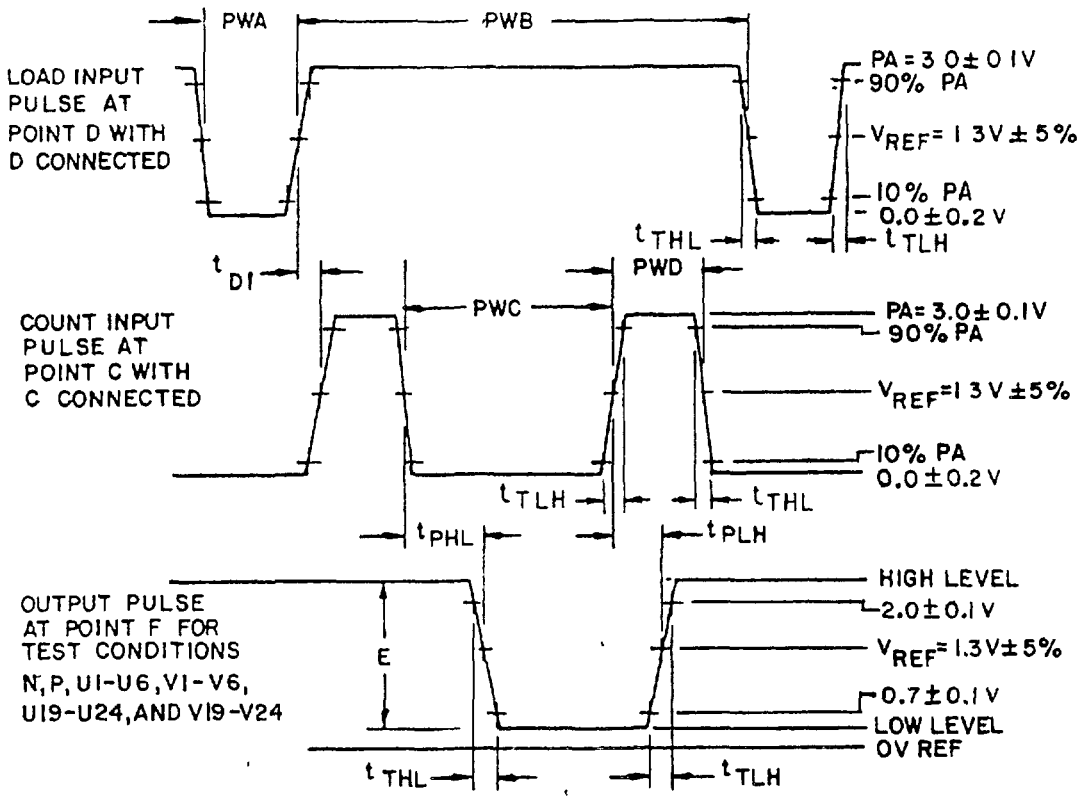
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load all outputs of circuit under test with load as specified on figure 11.

FIGURE 13. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{TLH} and t_{THL}) test circuit for tables XI, XII and XIII.

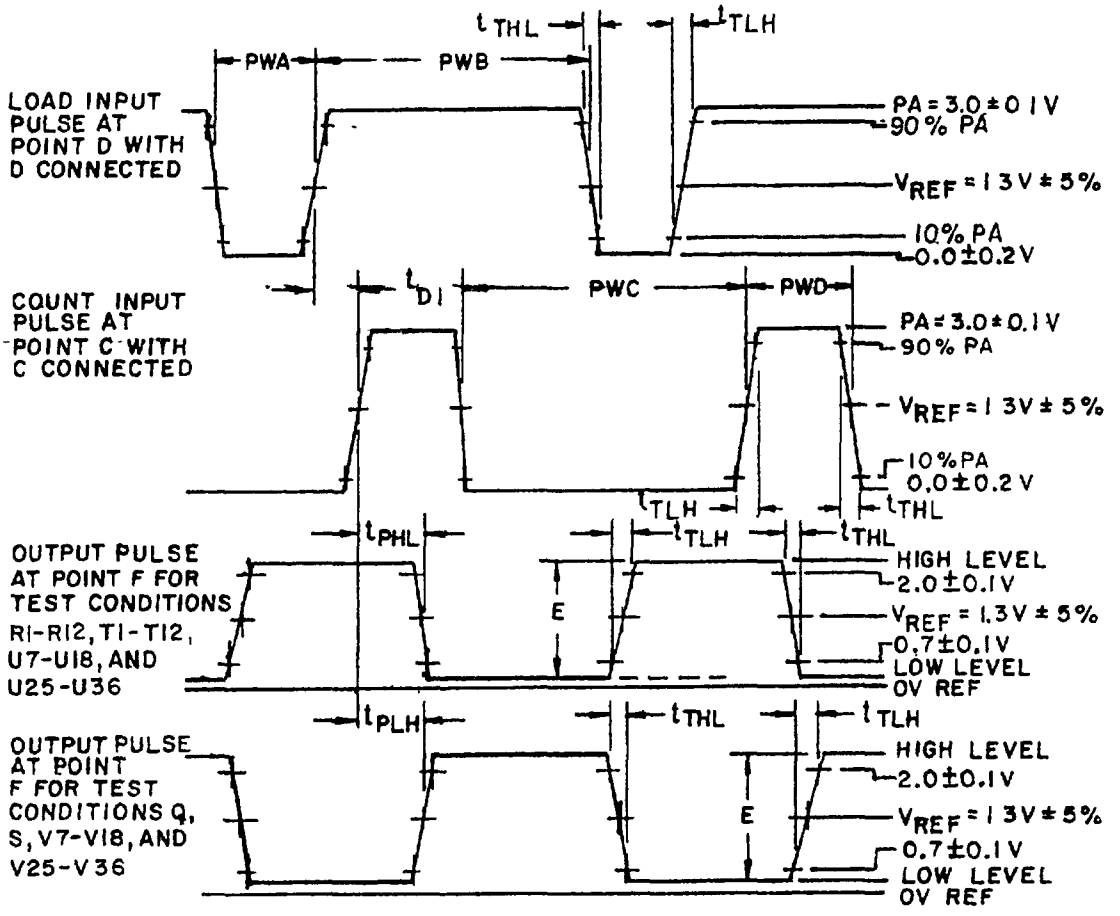
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NOTES.
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 1.5 ±0.1 µs.
   PWC = 750 ±75 ns.
   PWD = 250 ±25 ns.
   t_{D1} = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, verify that the low level is 0.3 volt maximum and the high level is 2.4 volts minimum.
3. All voltage levels are referenced to ground.
4. Waveform at point C is obtained by placing item J01-1 in the double pulse mode.

FIGURE 14. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XI and XIII.

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NOTES:
1. Input pulse parameters:
   PWA = 250 ±25 ns.
   PWB = 750 ±75 ns.
   PWC = 750 ±75 ns.
   PWD = 250 ±25 ns
   t_{D1} = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, verify that the low level is 0.3 volt maximum and the high level is 2.4 volts minimum.
3. All voltage levels are referenced to ground.

FIGURE 15. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XII and XIII.

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	L to 1QA (t_{PLH})	Propagation delay time, C_L = 50 pF	15,16, 23,40	Open	14	21	38	10	240	ns	16,17
X1	L to 1QA (t_{PHL})	↓	15,16, 23,40	↓	14	↓	38	↓	220	↓	↓
W2	L to 2QA (t_{PLH})	↓	8,9, 23,35	↓	13	↓	33	↓	240	↓	↓
X2	L to 2QA (t_{PHL})	↓	8,9, 23,35	↓	13	↓	33	↓	220	↓	↓
W3	L to 3QA (t_{PLH})	↓	2,3, 23,24	↓	5	↓	27	↓	240	↓	↓
X3	L to 1QB (t_{PHL})	↓	2,3, 23,24	↓	5	↓	27	↓	220	↓	↓
W4	L to 1QB (t_{PLH})	↓	14,15, 16,23	↓	40	↓	37	↓	240	↓	↓
X4	L to 1QB (t_{PHL})	↓	14,15, 16,23	↓	40	↓	37	↓	220	↓	↓

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W5	L to 1QA (t_{PLH}) Propagation delay time, C_L = 50 pF	8,9, 13,23	Open	35	21	34	10	240	ns	16,17
X5	L to 2QB (t_{PHL})	8,9, 13,23		35		34		220		
W6	L to 3QB (t_{PLH})	2,3, 5,23		24		28		240		
X6	L to 3QB (t_{PHL})	2,3, 5,23		24		28		220		
W7	L to 1QC (t_{PLH})	14,16, 23,40		15		18		240		
X7	L to 1QC (t_{PHL})	14,16, 23,40		15		18		220		
W8	L to 2QC (t_{PLH})	8,13, 23,25		9		30		240		
X8	L to 2QC (t_{PHL})	8,13, 23,25		9		30		220		

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W9	L to 3QC (t_{PLH}) Propagation delay time, C_L = 50 pF	2,5, 23,24	Open	3	21	6	10	240	ns	16,17
X9	L to 3QC (t_{PHL})	2,5, 23,24		3		6		220		
W10	L to 1QD (t_{PLH})	14,15, 23,40		16		19		240		
X10	L to 1QD (t_{PHL})	14,15, 23,40		16		19		220		
W11	L to 2QD (t_{PLH})	9,13, 23,35		8		29		240		
X11	L to 2QD (t_{PHL})	9,13, 23,35		8		29		220		
W12	L to 3QD (t_{PLH})	3,5, 23,24		2		25		240		
X12	L to 3QD (t_{PHL})	3,5, 23,24		2		25		220		

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W13	L to 1QA (t_{PLH}) Propagation delay time, C_L = 50 pF	15,16, 23,40	Open	14	21	38	10	255	ns	16,17
X13	L to 1QA (t_{PHL})	15,16, 23,40		14		38		235		
W14	L to 2QA (t_{PLH})	8,9, 23,35		13		33		255		
X14	L to 2QA (t_{PHL})	8,9, 23,35		13		33		235		
W15	L to 3QA (t_{PLH})	2,3, 23,24		5		27		255		
X15	L to 3QA (t_{PHL})	2,3, 23,24		5		27		235		
W16	L to 1QB (t_{PLH})	14,15, 16,23		40		37		255		
X16	L to 1QB (t_{PHL})	14,15, 16,23		40		37		235		

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W17	L to 2QB (t_{PLH}) Propagation delay time, C_L = 150 pF	8,9, 13,23	Open	35	21	34	10	255	ns	16,17
X17	L to 2QB (t_{PHL})	8,9, 13,23		35		34		235		
W18	L to 3QB (t_{PLH})	2,3, 5,23		24		28		255		
X18	L to 3QB (t_{PHL})	2,3, 5,23		24		28		235		
W19	L to 1QC (t_{PLH})	14,16, 23,40		15		18		255		
X19	L to 1QC (t_{PHL})	14,16, 23,40		15		18		235		
W20	L to 2QC (t_{PLH})	8,13, 23,35		9		30		255		
X20	L to 2QC (t_{PHL})	8,13, 23,35		9		30		235		

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W21	L to 3QC (t_{PLH}) Propagation delay time, C_L = 150 pF	2,5, 23,24	Open	3	21	6	10	255	ns	16,17
X21	L to 3QC (t_{PHL})	2,5, 23,24		3		6		235		
W22	L to 1QD (t_{PLH})	14,15, 23,40		16		19		255		
X22	L to 1QD (t_{PHL})	14,15, 23,40		16		19		235		
W23	L to 2QD (t_{PLH})	9,13, 23,35		8		29		255		
X23	L to 2QD (t_{PHL})	9,13, 23,35		8		29		235		
W24	L to 3QD (t_{PLH})	3,5, 23,24		2		25		255		
X24	L to 3QD (t_{PHL})	3,5, 23,24		2		25		235		

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y1	CL to 1QA	Propagation delay time, (t_{PHL}) C_L = 50 pF	15,16, 40	14	21	23	38	10	200	ns	16,18
Y2	CL to 2QA		8,9, 35	13			33				
Y3	CL to 3QA		2,3, 24	5			27				
Y4	CL to 1QB		14,15, 16	40			37				
Y5	CL to 2QB		8,9, 13	35			34				
Y6	CL to 3QB		2,3, 5	24			28				
Y7	CL to 1QC		14,16, 40	15			18				
Y8	CL to 2QC		8,13, 35	9			30				
Y9	CL to 3QC		2,5, 24	3			6				
Y10	CL to 1QD		14,15, 40	16			19				
Y11	CL to 2QD		9,13, 35	8			29				

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

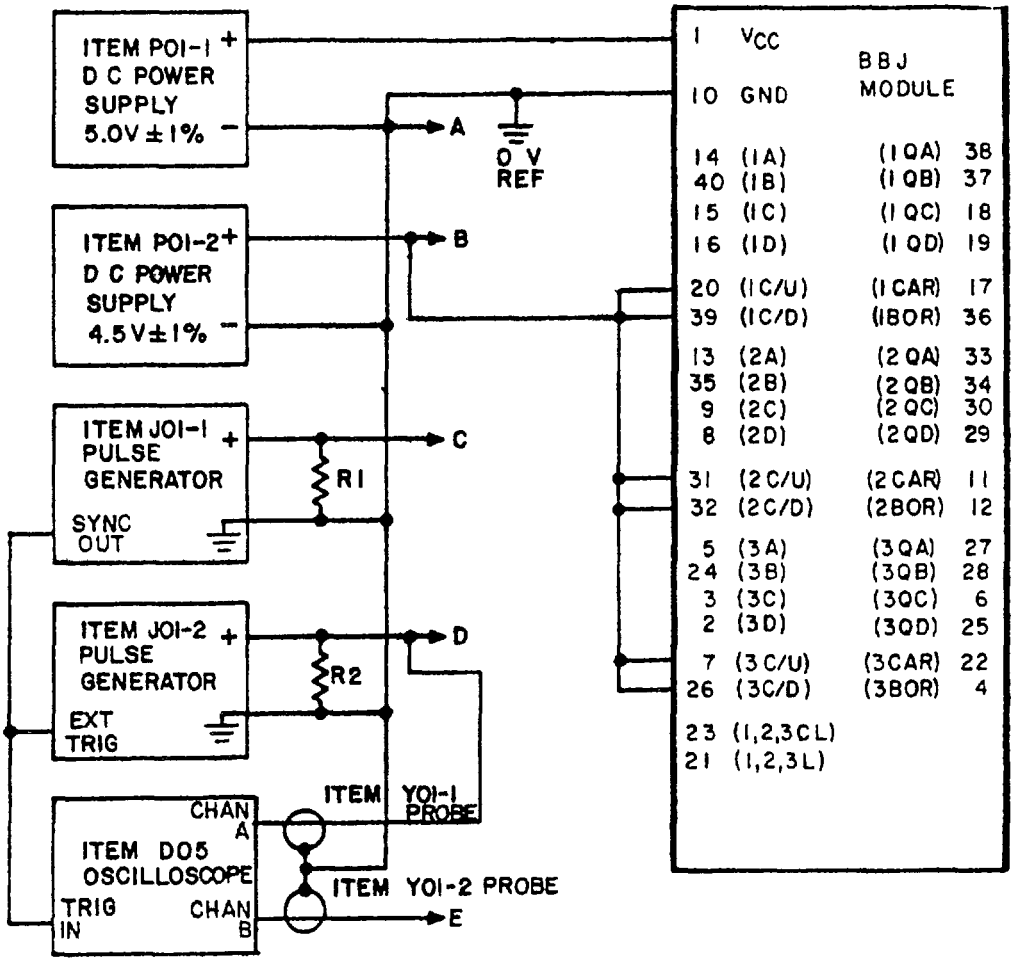
Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y12	CL to 3QD	Propagation delay time, (t_{PHL}) C_L = 50 pF	3,5, 24	2	21	23	25	10	200	ns	16,18
Y13	CL to 1QA	Propagation delay time, (t_{PHL}) C_L = 150 pF	15,16, 40	14			38		215		
Y14	CL to 2QA		8,9, 35	13			33				
Y15	CL to 3QA		2,3, 24	5			27				
Y16	CL to 1QB		14,15, 16	40			37				
Y17	CL to 2QB		8,9, 13	35			34				
Y18	CL to 3QB		2,3, 5	24			28				
Y19	CL to 1QC		14,16, 40	15			13				
Y20	CL to 2QC		8,13, 35	9			30				
Y21	CL to 3QC		2,5, 24	3			6				
Y22	CL to 1QD		14,15, 40	16			19				

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Y23	CL to 2QD	Propagation delay time, (t_{PHL}) C_L = 150 pF	9,13, 35	8	21	23	29	10	215	ns	16,18
Y24	CL to 3QD	Propagation delay time, (t_{PHL}) C_L = 150 pF	3,5, 24	2	21	23	25	10	215	ns	16,18

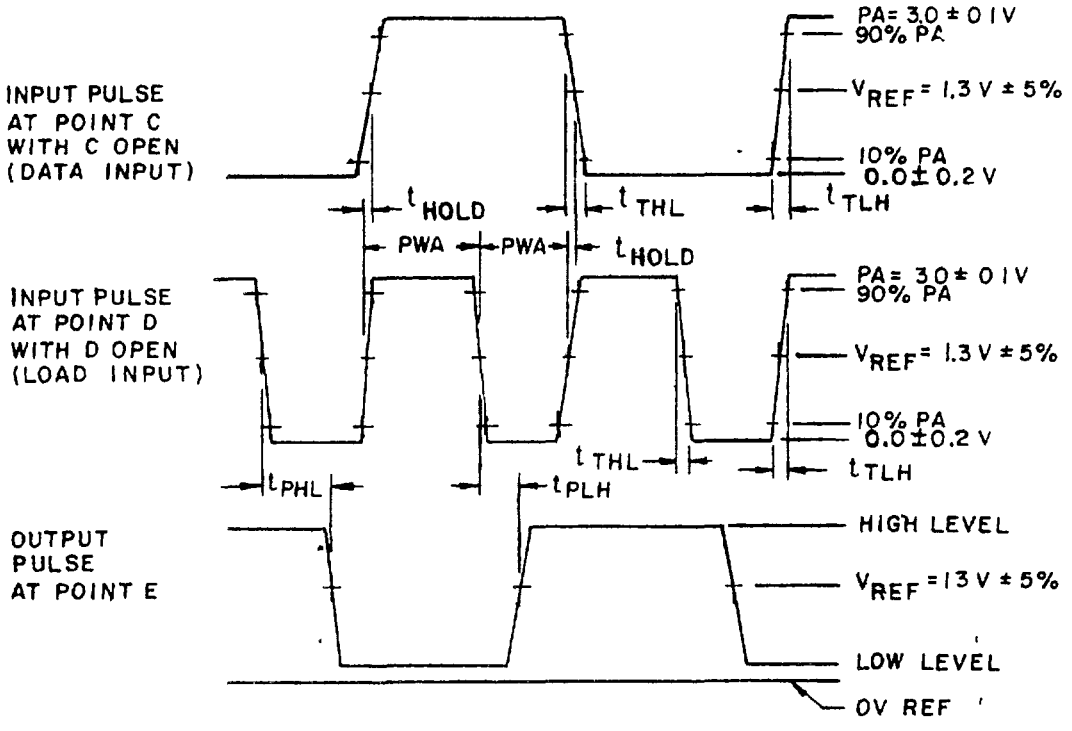
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load all outputs of circuit under test with load as specified on figure 11.

FIGURE 16. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV.

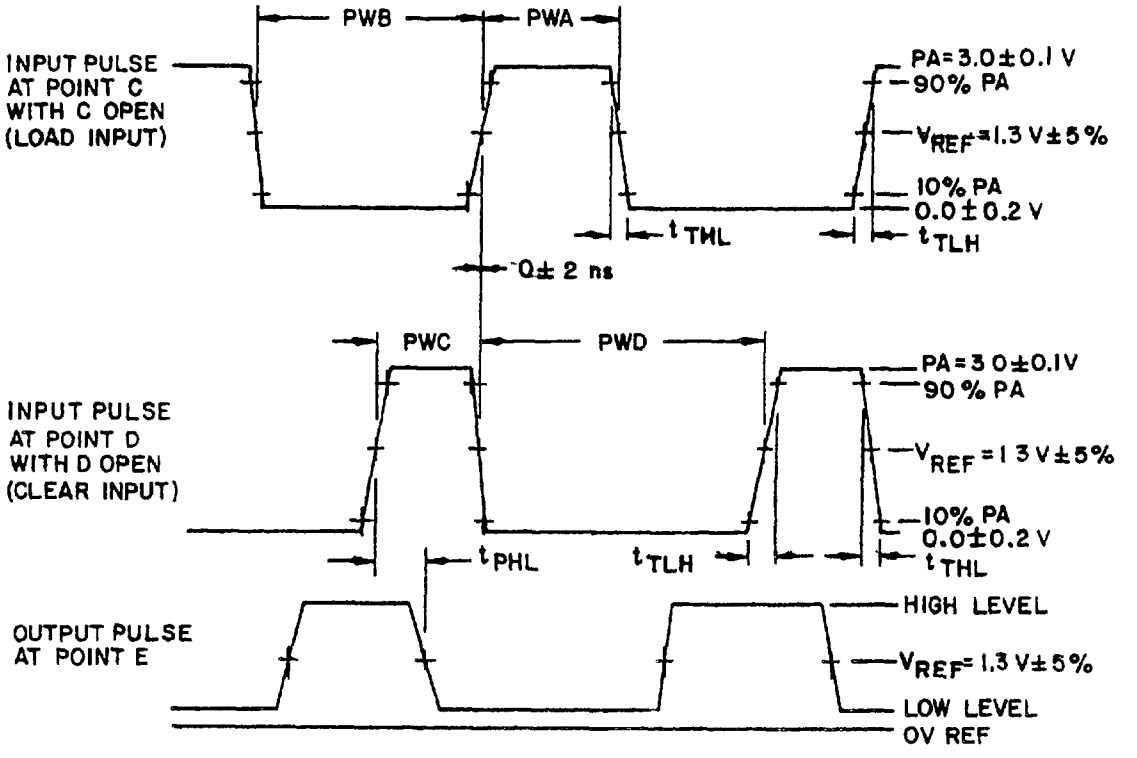
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NOTES.
1. Input pulse parameters.
   PWA = 500 ±50 ns.
   $t_{HOLD} = 0\ ^{+2}_{-0}$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Verify that the output low level is 0.3 volt maximum and the output high level is 2.4 volts minimum.
3. All voltage levels are referenced to ground.

FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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NOTES:
1. Input pulse parameters
   PWA = 300 ±30 ns.
   PWB = 700 ±70 ns.
   PWC = 200 $^{+5}_{-0}$ ns (clear pulse width).
   PWD = 800 ±80 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify that the output low level is 0.3 volt maximum and the output high level is 2.4 volts minimum.
3. All voltage levels are referenced to ground.

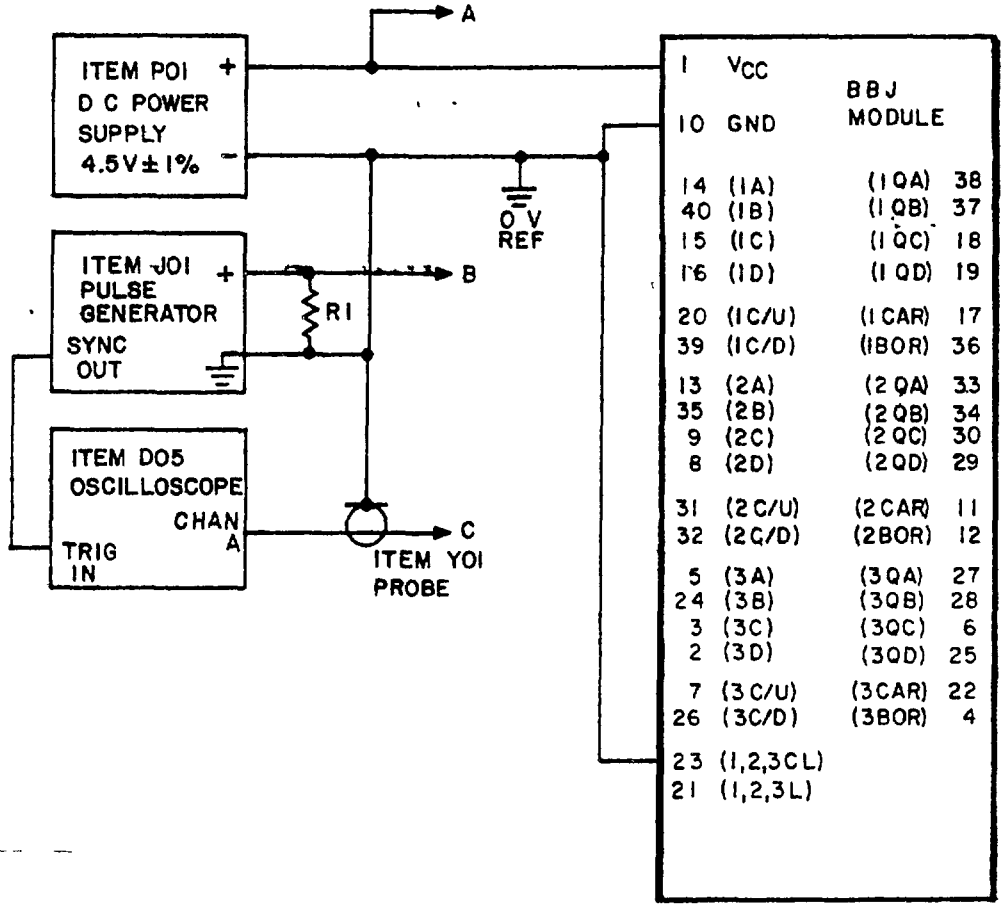
FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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TABLE XV. Maximum counting frequency check (carry and borrow period) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	Maximum counting frequency check, carry period (circuit 1)	39,21	20	17	6.3	6.6	µs	19,20
Z2	Maximum counting frequency check, carry period (circuit 2)	32,21	31	11				
Z3	Maximum counting frequency check, carry period (circuit 3)	26,31	7	22				
Z4	Maximum counting frequency check, borrow period (circuit 1)	20,21	39	36				
Z5	Maximum counting frequency check, borrow period (circuit 2)	31,21	32	12				
Z6	Maximum counting frequency check, borrow period (circuit 3)	7,21	26	4				

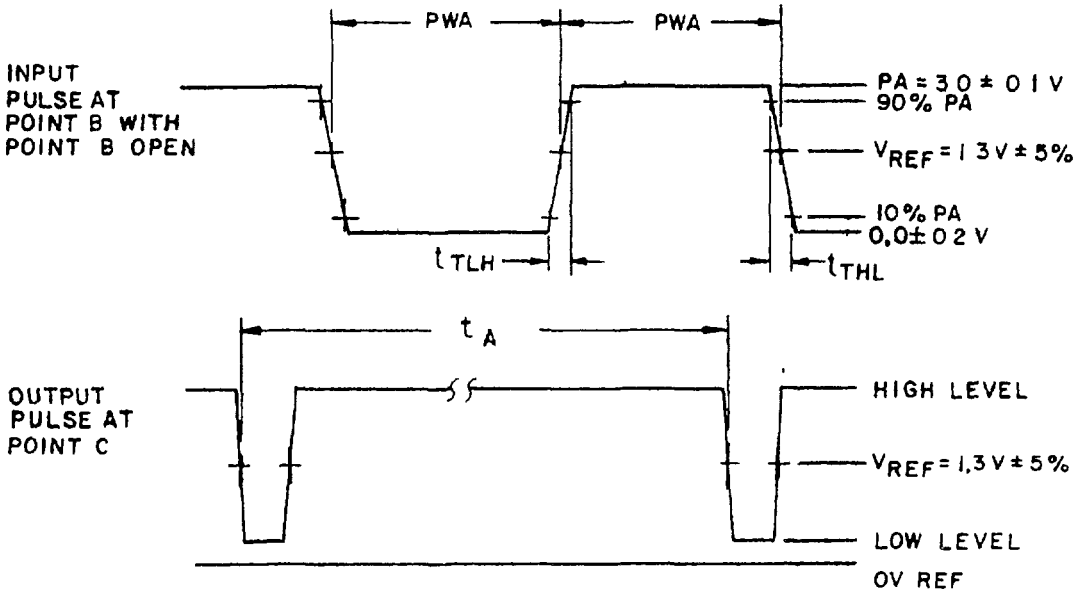
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load all outputs of circuit under test with the load as specified in figure 11.

FIGURE 19. Maximum counting frequency check (carry and borrow period) test circuit for table XV.

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NOTES:
1. Input pulse parameters:

   PWA = 200 $^{+5}_{-0}$ ns.

   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Verify that output low level is 0.3 volt maximum and that the high level is 2.4 volts minimum.
3. All voltage levels are referenced to ground.
4. t_A = Carry or borrow period.

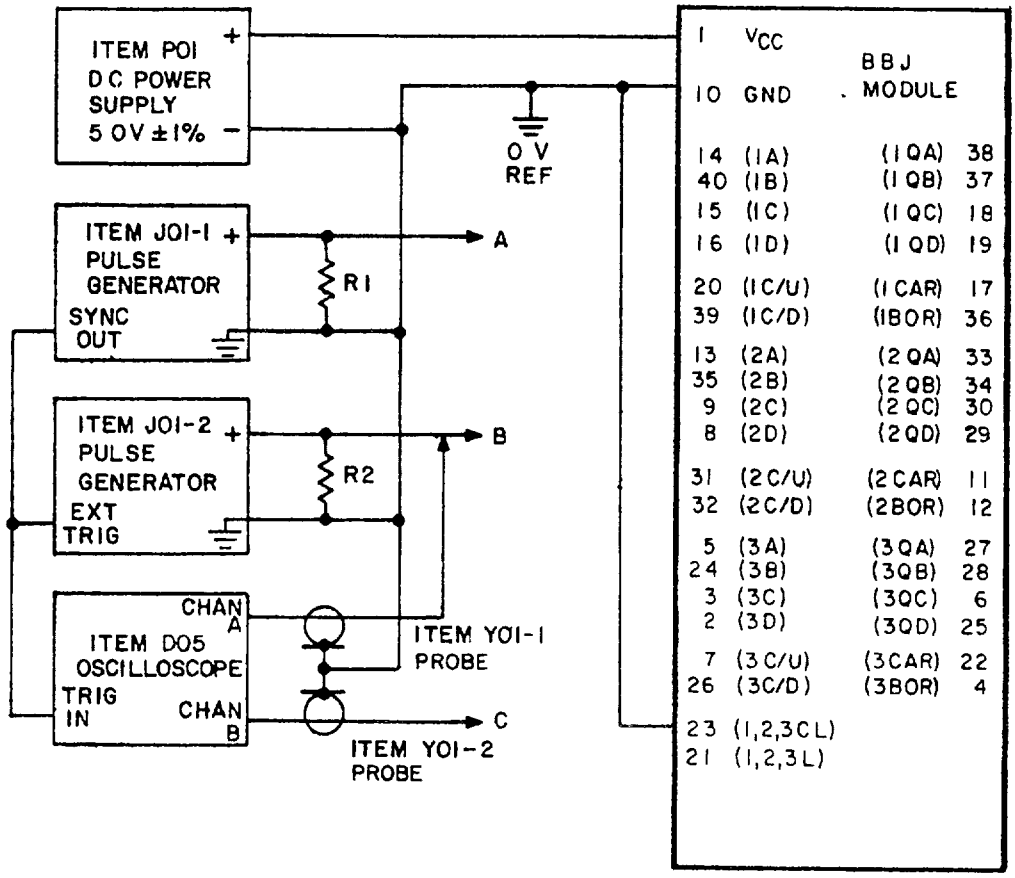
FIGURE 20. Maximum counting frequency check (carry and borrow period) test circuit waveforms for table XV.

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TABLE XVI. Setup time (data to load) and minimum load pulse width check test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
AA1	1QA Setup time (data to load) and minimum load pulse width check	14	21	38	935	1,070	ns	21,22
AA2	2QA	13		33				
AA3	3QA	5		27				
AA4	1QB	40		37				
AA5	2QB	35		34				
AA6	3QB	24		28				
AA7	1QC	15		18				
AA8	2QC	9		30				
AA9	3QC	3		6				
AA10	1QD	16		19				
AA11	2QD	8		29				
AA12	3QD	2		25				

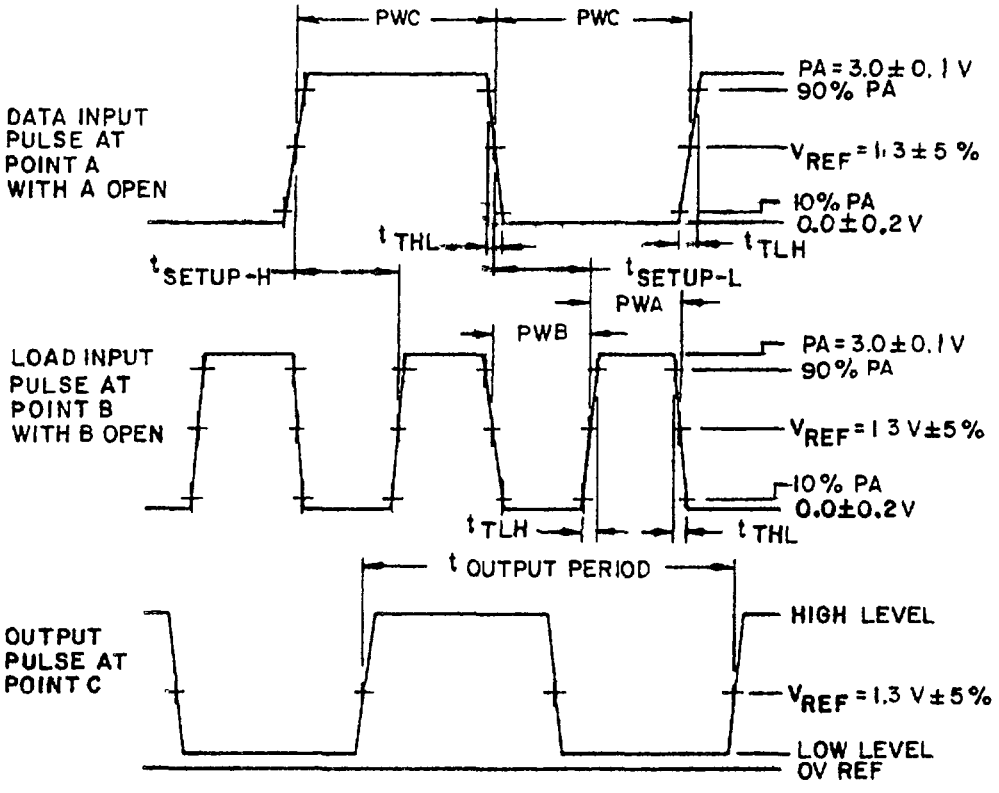
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NOTES
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load all outputs of circuit under test with the load as specified in figure 11.

FIGURE 21. Setup time (data to load) and minimum load pulse width check test circuit for table XVI.

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NOTES:
1. Input pulse parameters:

   PWA = 200 $^{+5}_{-0}$ ns.
   PWB = 300 ±30 ns.
   $t_{SETUP-L} = t_{SETUP-H} = 110\ ^{+5}_{-0}$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWC = 500 $^{+5}_{-0}$ ns.

2. All voltage levels are referenced to ground.
3. Verify that output low level is 0.3 volt maximum and that output high level is 2.4 volts minimum.

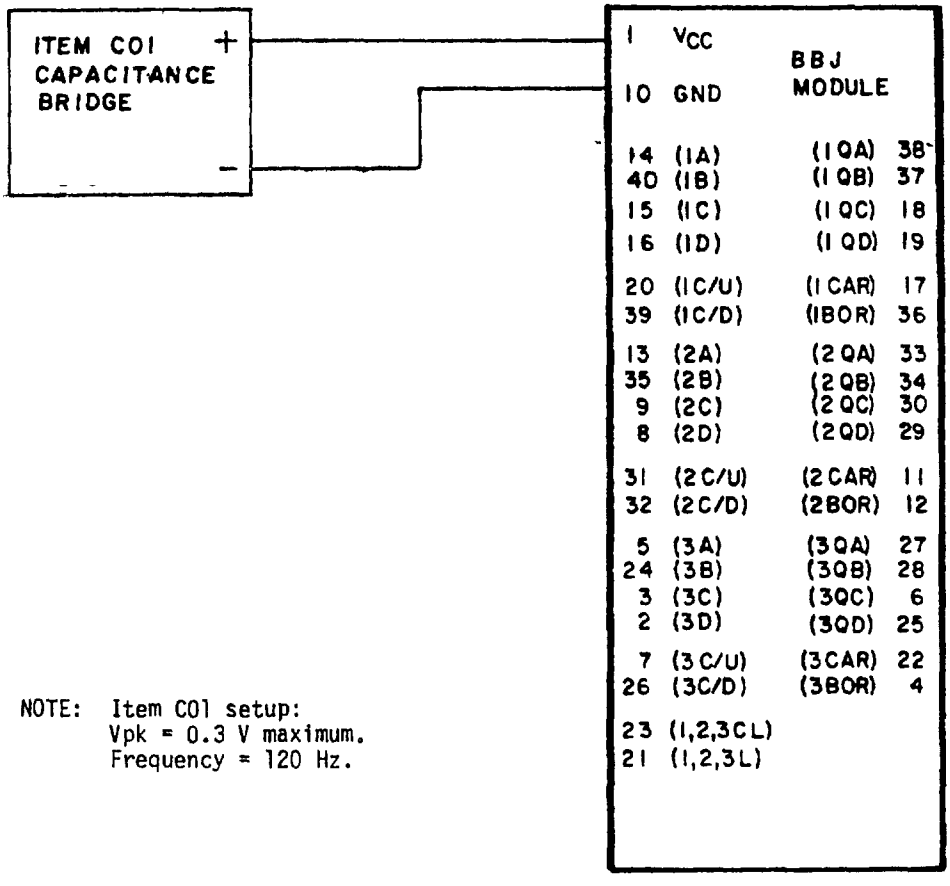
FIGURE 22. Setup time (data to load) and minimum load pulse width check test circuit waveforms for table XVI.

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TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
BB	Power supply filter capacitance	1.6	3.0	µF	23

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NOTE: Item C01 setup:
Vpk = 0.3 V maximum.
Frequency = 120 Hz.

FIGURE 23. Power supply filter capacitance test circuit for table XVII.

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TABLE XVIII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC1	1QA	Output crosstalk, high level	14	Open	21	2,3,5,8,9, 13,15,16, 24,35,40	38	0	400	mV	24,25
CC2	1QB		40			2,3,5,8,9, 13,14,15, 16,24,35	37				
CC3	1QC		15			2,3,5,8,9, 13,14,16, 24,35,40	18				
CC4	1QD		16			2,3,5,8,9, 13,14,15, 24,35,40	19				
CC5	2QA		13			2,3,5,8,9, 14,15,16, 24,35,40	33				
CC6	2QB		35			2,3,5,8,9, 13,14,15, 16,24,40	34				
CC7	2QC		9			2,3,5,8, 13,14,15, 16,24,35, 40	30				
CC8	2QD		8			2,3,5,9, 13,14,15, 16,24,35, 40	29				

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TABLE XVIII. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
	-		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC9	3OA	Output crosstalk, high level	5	Open	21	2,3,8,9, 13,14,15, 16,24,35, 40	27	0	400	mV	24,25
CC10	3QB		24			2,3,5,8, 9,13,14, 15,16,35, 40	28				
CC11	3QC		3			2,5,8,9, 13,14,15, 16,24,35, 40	6				
CC12	3QD		2			3,5,8,9, 13,14,15, 16,24,35, 40	25				
DD1	1QA	Output crosstalk, low level	Open	14		2,3,5,8, 9,13,15, 16,24,35, 40	38				
DD2	1QB	Output crosstalk, low level	Open	40		2,3,5,8, 9,13,14, 15,16,24, 35	37				
DD3	1QC	Output crosstalk, low level	Open	15		2,3,5,8, 9,13,14, 15,24,35, 40	18				

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TABLE XVIII. Output crosstalk (high-level and low-level) test - Continued.

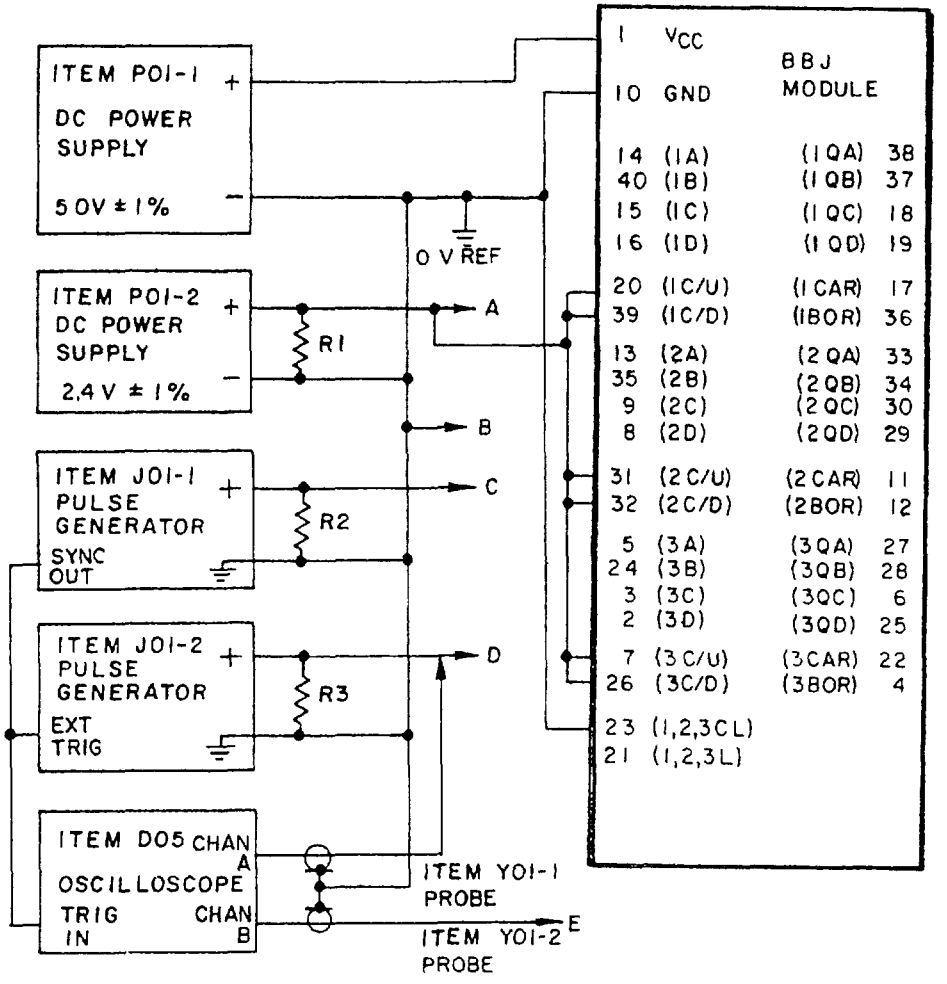
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
DD4	1QD Output crosstalk, low level	Open	16	21	2,3,5,8, 9,13,14, 15,24,35, 40	19	0	400	mV	24,25
DD5	2QA		13		2,3,5,8, 9,14,15, 16,24,35, 40	33				
DD6	2QB		35		2,3,5,8, 9,13,14, 15,16,24, 40	34				
DD7	2QC		9		2,3,5,8, 13,14,15, 16,24,35, 40	30				
DD8	2QD		8		2,3,5,9, 13,14,15, 16,24,35, 40	29				
DD9	3QA		5		2,3,8,9, 13,14,15, 16,24,35, 40	27				
DD10	3QB		24		2,3,5,8, 9,13,14, 15,16,35, 40	28				

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TABLE XVIII. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
DD11	3QC Output crosstalk, low level	Open	3	21	2,5,8,9, 13,14,15, 16,24,35, 40	6	0	400	mV	24,25
DD12	3QD Output crosstalk, low level	Open	2	21	3,5,8,9, 13,14,15, 16,24,35, 40	25	0	400	mV	24,25

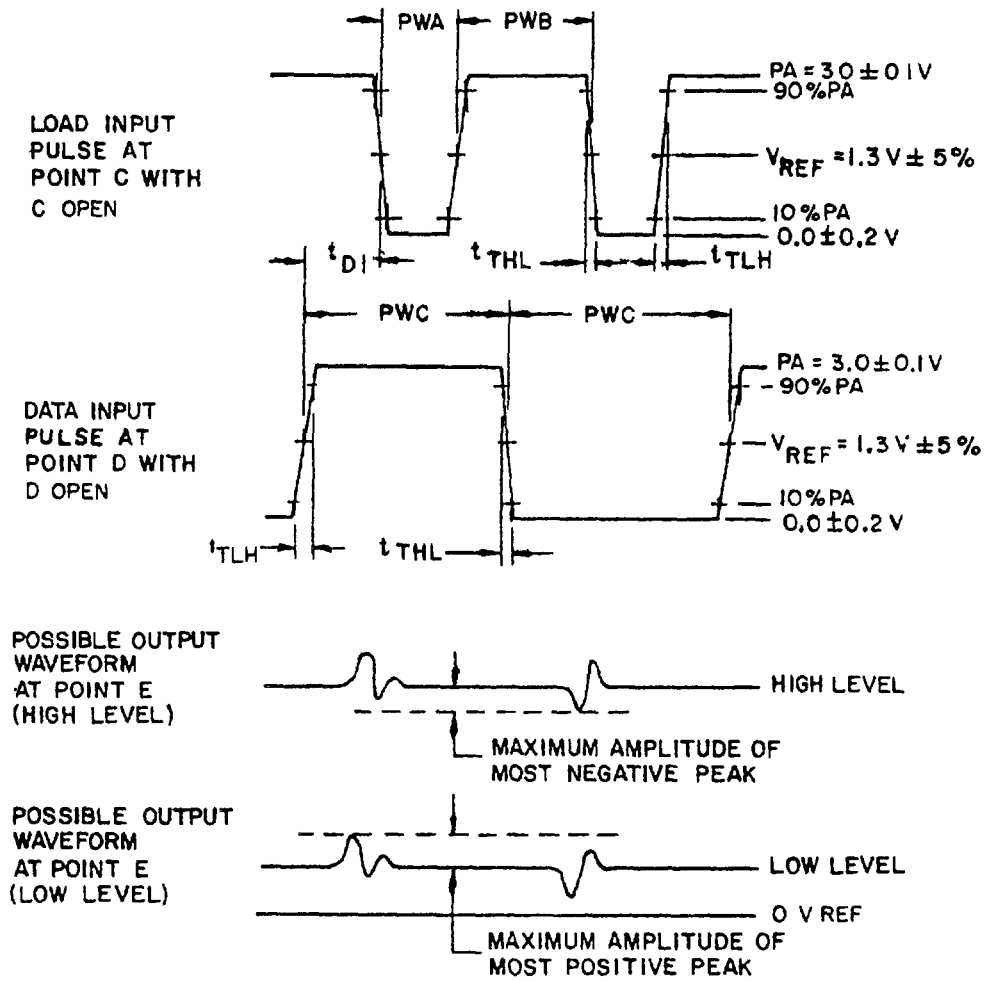
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NOTES:
1. R1-R3 = 50Ω ±5%, 1 W.
2. Load all outputs with load as specified on figure 11.

FIGURE 24. Output crosstalk (high-level and low-level) test circuit for table XVIII.

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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
PWB = 500 ±50 ns.
PWC = 1,000 ±100 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
t_{D1} = 250 ±25 ns.

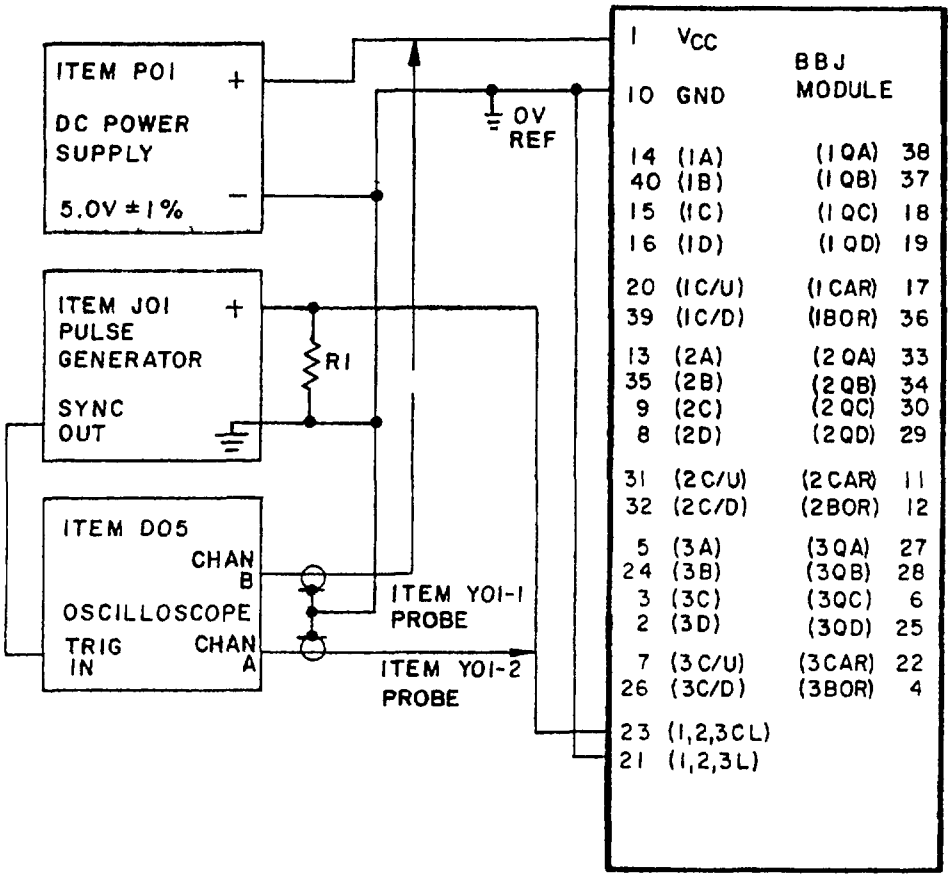
FIGURE 25. Output crosstalk (high-level and low-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
EE	Power supply transient amplitude	0	400	mV	26,27

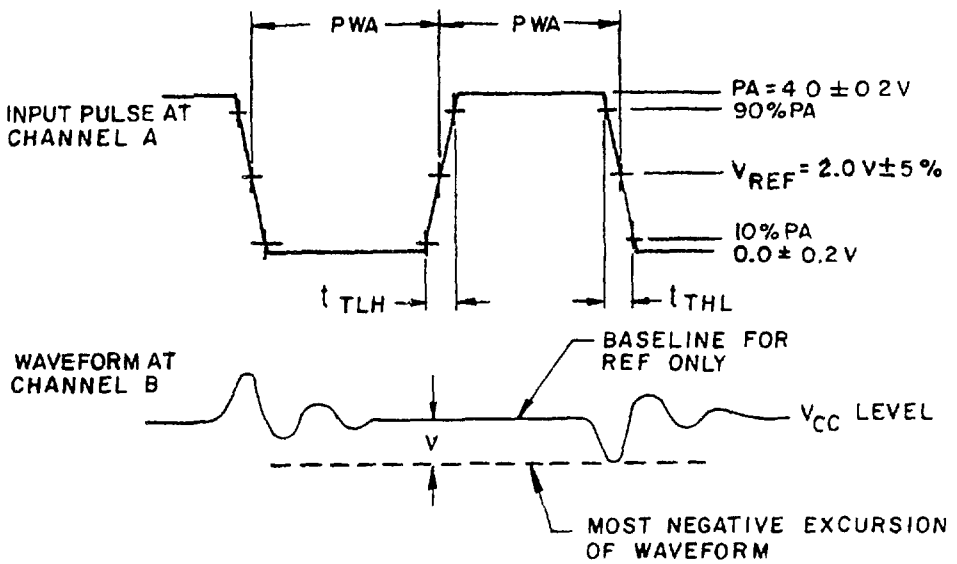
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NOTE: R1 = 50Ω ±5%, 1 W.

FIGURE 26. Power supply transient amplitude test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. V = Measurement is made from the baseline to the most negative excursion of waveform.

FIGURE 27. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Input capacitance test.

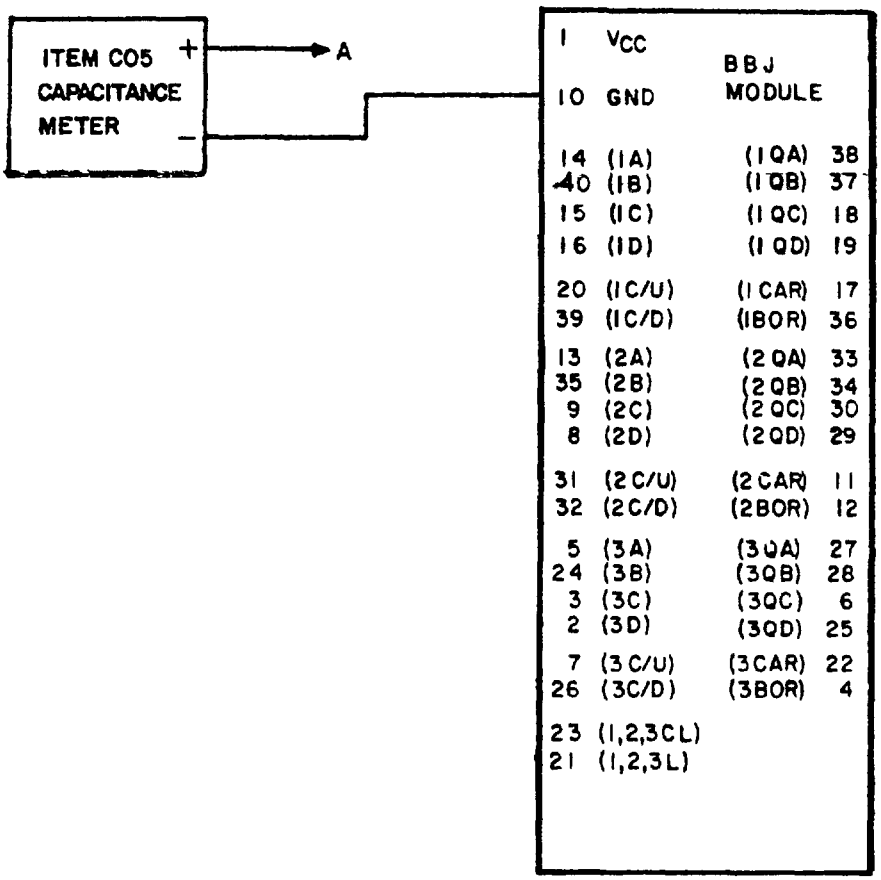
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
FF1	1A Input capacitance	14	2	12	pF	28
FF2	1B	40				
FF3	1C	15				
FF4	1D	16				
FF5	1C/U	20				
FF6	1C/D	39				
FF7	2A	13				
FF8	2B	35				
FF9	2C	9				
FF10	2D	8				
FF11	2C/U	31				
FF12	2C/D	32				
FF13	3A	5				
FF14	3B	24				
FF15	3C	3				
FF16	3D	2				
FF17	3C/U	7				
FF18	3C/D	26				

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
FF19	1,2,3CL	Input capacitance	23	6	36	pF	28
FF20	1,2,3L	Input capacitance	21	6	36	pF	28

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 28. Input capacitance test circuit for table XX.

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